Strong Duality and Saddle Points

Chapter 18 showed how minimizing a Lagrangian produces a global lower bound and how a matching bound certifies a primal solution. We now ask when such a bound is guaranteed to exist. We will first derive the role of Slater's condition in convex strong duality, use examples to separate its hypotheses from attainment, and then examine the stronger conclusion available for linear programs. Matching values will lead us to KKT conditions and global saddle points. Finally, we will use that geometry to investigate primal–dual iterations: why naive gradient updates can fail, and how ALM introduces a moving regularization that gives both a stable dual step and a smooth dual objective.

Learning objectives

By the end of this chapter, you should be able to:

- express the primal and dual as an infimum–supremum pair;
- apply Slater's theorem while distinguishing zero gap from attainment;
- derive a linear-programming dual and its bidual, and justify LP strong duality;
- connect matching values, convex KKT conditions, and primal recovery;
- prove the equivalence between a global saddle point and attained zero gap;
- explain why naive primal–dual gradient updates can fail; and
- derive the proximal and smooth-envelope interpretations of convex ALM, distinguishing exact identities from inexact implementation.

19.1 The question left by weak duality

From Chapter 18:

$$p^* = \inf\{f(\boldsymbol{x}) : \boldsymbol{x} \in \mathbb{R}^n,\ \boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{0},\ \boldsymbol{h}(\boldsymbol{x}) \le \boldsymbol{0}\}, \qquad d^* = \sup_{\boldsymbol{\lambda}\in\mathbb{R}^m,\,\boldsymbol{\nu}\ge\boldsymbol{0}} \theta(\boldsymbol{\lambda},\boldsymbol{\nu}),$$

where all problem functions are finite and real-valued on $\mathbb{R}^n$, and

$$L(\boldsymbol{x},\boldsymbol{\lambda},\boldsymbol{\nu}) = f(\boldsymbol{x}) + \boldsymbol{\lambda}^\mathsf{T}\boldsymbol{g}(\boldsymbol{x}) + \boldsymbol{\nu}^\mathsf{T}\boldsymbol{h}(\boldsymbol{x}), \qquad \theta(\boldsymbol{\lambda},\boldsymbol{\nu}) = \inf_{\boldsymbol{x}\in\mathbb{R}^n} L(\boldsymbol{x},\boldsymbol{\lambda},\boldsymbol{\nu}).$$

Weak duality, proved in Theorem 18.1, gives the chain

$$\theta(\boldsymbol{\lambda},\boldsymbol{\nu}) \le d^* \le p^* \le f(\boldsymbol{x}) \tag{19.1}$$

for any primal–dual feasible pair. When both optimal values are finite, strong duality means $d^* = p^*$; it does not assert that either optimal value is attained. The optimal gap $p^* - d^*$ must also be distinguished from the computable gap $f(\boldsymbol{x}) - \theta(\boldsymbol{\lambda},\boldsymbol{\nu})$ of a candidate pair. With infinite values the order statement $d^* \le p^*$ remains meaningful, whereas subtraction can be undefined.

To see what equality of the two optimal values asks for, first fix $\boldsymbol{x}$ and maximize its Lagrangian over the allowed multipliers. If $\boldsymbol{x}$ is feasible, the equality term vanishes and every inequality term is nonpositive; choosing $\boldsymbol{\nu} = \mathbf{0}$ attains the maximum $f(\boldsymbol{x})$. If some $g_i(\boldsymbol{x}) \neq 0$, the unrestricted multiplier λ_i can make its term arbitrarily large. If some $h_j(\boldsymbol{x}) > 0$, the same is true by taking $\nu_j \to +\infty$. Consequently,

$$\sup_{\boldsymbol{\lambda}\in\mathbb{R}^m,\,\boldsymbol{\nu}\geq\mathbf{0}} L(\boldsymbol{x},\boldsymbol{\lambda},\boldsymbol{\nu}) = \begin{cases} f(\boldsymbol{x}), & \boldsymbol{g}(\boldsymbol{x}) = \mathbf{0},\ \boldsymbol{h}(\boldsymbol{x}) \leq \mathbf{0}, \\ +\infty, & \text{otherwise.} \end{cases}$$

Taking the infimum over $\boldsymbol{x}$ therefore recovers the original problem:

$$p^* = \inf_{\boldsymbol{x}\in\mathbb{R}^n} \sup_{\boldsymbol{\lambda},\,\boldsymbol{\nu}\geq\mathbf{0}} L(\boldsymbol{x},\boldsymbol{\lambda},\boldsymbol{\nu}), \qquad d^* = \sup_{\boldsymbol{\lambda},\,\boldsymbol{\nu}\geq\mathbf{0}} \inf_{\boldsymbol{x}\in\mathbb{R}^n} L(\boldsymbol{x},\boldsymbol{\lambda},\boldsymbol{\nu}). \tag{19.2}$$

Thus strong duality asks when these two orders give the same value. The general inequality $\sup\inf \leq \inf\sup$ gives weak duality, but it cannot justify interchanging the operations. We first introduce a convexity and strict-feasibility condition under which a separating hyperplane supplies the missing lower bound.

19.2 Convex strong duality and Slater's condition

19.2.1 The guarantee and its geometric mechanism

Consider a problem with a finite convex objective and finite convex inequality functions:

$$\begin{aligned} p^* = \inf_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & \boldsymbol{B}\boldsymbol{x} = \boldsymbol{b}, \\ & \boldsymbol{h}(\boldsymbol{x}) \leq \mathbf{0}. \end{aligned} \tag{19.3}$$

The equality constraints are affine. Convexity alone will not ensure that the best Lagrangian lower bound reaches p^*. We impose a strict-feasibility condition to rule out a particular obstruction.

Definition 19.1 (Slater's condition)**.** The strict form of Slater's condition holds if there is $\widehat{\boldsymbol{x}} \in \mathbb{R}^n$ with

$$\boldsymbol{B}\widehat{\boldsymbol{x}} = \boldsymbol{b}, \qquad h_j(\widehat{\boldsymbol{x}}) < 0 \quad (j = 1, \ldots, p).$$

When there are no inequalities, this requires only equality feasibility.

Theorem 19.1 (Convex strong duality)**.** *Suppose $f, h_1, \ldots, h_p$ are finite convex functions on $\mathbb{R}^n$, the equality constraints are affine, $p^* \in \mathbb{R}$, and Slater's condition holds. Then $d^* = p^*$, and some dual-feasible $(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ attains d^*.*

We describe the geometry before giving the full proof in Section 19.8. Record an equality residual $\boldsymbol{u}$, an upper bound $\boldsymbol{v}$ on the inequality residuals, and an upper bound t on the objective. Their achievable strict upper set is

$$\mathcal{O} = \{(\boldsymbol{u}, \boldsymbol{v}, t) : \exists \boldsymbol{x}, \quad \boldsymbol{B}\boldsymbol{x} - \boldsymbol{b} = \boldsymbol{u}, \quad \boldsymbol{h}(\boldsymbol{x}) < \boldsymbol{v}, \quad f(\boldsymbol{x}) < t\}.$$

Affinity and convexity make this a convex set. After redundant equalities are removed, it is open. The point $(\mathbf{0}, \mathbf{0}, p^*)$ lies outside it: a point witnessing membership would be feasible with objective below p^*. A separating hyperplane therefore supplies coefficients with

$$\bar{\boldsymbol{\lambda}}^{\mathsf{T}}\boldsymbol{u} + \bar{\boldsymbol{\nu}}^{\mathsf{T}}\boldsymbol{v} + \eta t \geq \eta p^*.$$

Because the upper coordinates can increase freely, $\bar{\boldsymbol{\nu}} \geq \mathbf{0}$ and $\eta \geq 0$. Slater's strict feasible point rules out $\eta = 0$, as we can take $\boldsymbol{u} = \mathbf{0}$ and $\boldsymbol{v} < \mathbf{0}$. We may consequently divide by η and obtain multipliers satisfying

$$L(\boldsymbol{x}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) \geq p^* \quad \text{for every } \boldsymbol{x} \in \mathbb{R}^n.$$

The global lower bound reaches p^*, while weak duality prevents it from exceeding p^*. This is the source of both zero gap and dual attainment. The full proof checks openness, the positive objective coefficient, and the mapping back from reduced equality coordinates.

Neither differentiability nor linear independence of the original equality constraints is required. Primal attainment is a separate issue: the theorem supplies an optimal multiplier even if the primal infimum is approached only along an unbounded sequence. Also, strict Slater is sufficient rather than necessary. The linear-programming specialization below will give zero-gap problems without strict feasibility of all displayed inequalities.

To visualize this supporting geometry and a possible obstruction (see Figure 19.1), consider a single inequality and define $v(u) = \inf\{f(\boldsymbol{x}) : h(\boldsymbol{x}) \leq u\}$, with value $+\infty$ when infeasible. A multiplier attaining zero gap would give the global affine bound $v(u) \geq v(0) - \nu^* u$. For $\min e^{-x}$ subject to $x \leq 0$, perturbing the constraint to $x \leq u$ gives $v(u) = e^{-u}$, supported at $u = 0$ by $1 - u$. The right panel depicts a different convex value function, with $v(0) = 1$ and $v(u) = 0$ for every $u > 0$. A concrete problem producing this function is developed in Example 19.2. No finite ν can make $0 \geq 1 - \nu u$ hold for every positive u. Only a vertical separating line remains, which has zero objective coefficient and cannot be normalized into the desired Lagrangian bound.

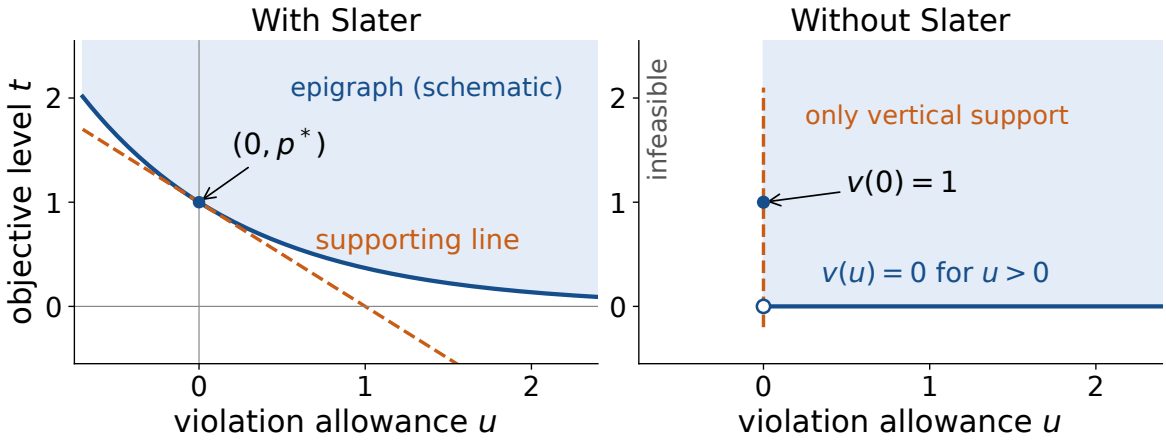


Figure 19.1: Scalar perturbation geometry. On the left, a finite supporting slope gives the multiplier in a lower bound touching p^*. On the right, the jump in the convex positive-gap example prevents such a support; the vertical separator cannot provide a finite multiplier certificate.

19.3 Hypotheses, counterexamples, and attainment

To understand exactly what Slater's theorem asserts, first remove convexity while retaining strict feasibility, and then reverse these choices. After checking the resulting gaps, we will examine what can be concluded about optimizers rather than values. All restrictions remain explicit constraints on the full Euclidean space.

Example 19.1 (Strict feasibility without convexity)**.** Let $q(x) = (x^2 - 1)^2$ and consider

$$\min_{x \in \mathbb{R}} q(x) \quad \text{subject to} \quad x = 0, \quad x^2 - 1 \leq 0.$$

The only feasible point is $x = 0$, where the inequality is strict and $p^* = 1$. For unrestricted λ and $\nu \geq 0$,

$$L(x, \lambda, \nu) = q(x) + \lambda x + \nu(x^2 - 1).$$

Evaluating at $x = 1$ and $x = -1$ eliminates both q and the inequality term, so $\theta(\lambda, \nu) \leq -|\lambda| \leq 0$. At $(\lambda, \nu) = (0, 0)$ the infimum is 0, hence $d^* = 0 < p^* = 1$. The missing hypothesis is convexity of q on $\mathbb{R}$. In particular, the triple $(0, 0, 0)$ satisfies KKT, since $q'(0) = 0$, but its dual value does not match the primal objective.

We next retain convexity and remove strict feasibility. The obstruction will appear in the perturbation value, even though the data can be made continuously differentiable.

Example 19.2 (Convexity without strict feasibility)**.** Set $a(x, y) = \sqrt{x^2 + y^2} - y \geq 0$ and consider

$$\min_{(x,y) \in \mathbb{R}^2} e^{-x} \quad \text{subject to} \quad h(x, y) = a(x, y)^2 \leq 0.$$

The norm-minus-linear function a is convex and nonnegative. Squaring is convex and increasing on $[0, \infty)$, so h is convex. Its zero set is $\{(0, y) : y \geq 0\}$; therefore $p^* = 1$ and strict feasibility is impossible. Both terms in $L = e^{-x} + \nu h(x, y)$ are nonnegative. To determine its infimum for any fixed $\nu \geq 0$, take $(x, y) = (t, t^3)$ and rationalize:

$$a(t, t^3) = \frac{t^2}{\sqrt{t^2 + t^6} + t^3} \leq \frac{1}{2t}, \qquad 0 \leq L(t, t^3, \nu) \leq e^{-t} + \frac{\nu}{4t^2} \longrightarrow 0.$$

Thus $\theta(\nu) = 0$ for every $\nu \geq 0$, and $d^* = 0 < 1 = p^*$. Every multiplier is dual optimal, although no inner infimum is attained. This example is continuously differentiable as well as convex: away from the origin the formula is smooth, while $h(x, y) \leq 4(x^2 + y^2)$ and $\|\nabla h(x, y)\| \leq 8 \|(x, y)\|$ show differentiability with zero gradient and gradient continuity at the origin. The same construction with the unsquared constraint $a \leq 0$ also has gap 1; squaring shows that nonsmoothness is not the cause of the gap.
For the perturbation value in Figure 19.1, we have $v(0) = 1$. For each fixed $u > 0$, the sequence (t, t^3) eventually satisfies $h(t, t^3) \leq u$ while its objective approaches zero. Hence $v(u) = 0$; for $u < 0$ the perturbed problem is infeasible because $h \geq 0$.

These counterexamples prevent us from discarding either hypothesis in a general guarantee. They do not say that every problem violating a hypothesis must have a positive gap. We can test this distinction with the same quartic.

Example 19.3 (Neither hypothesis is necessary for a particular zero-gap problem)**.** With the same nonconvex objective q, consider

$$\min_{x \in \mathbb{R}} q(x) \quad \text{subject to} \quad x^2 \leq 0.$$

Again the only feasible point is 0 and $p^* = 1$, but now strict Slater also fails. Choosing $\nu = 2$ gives $L(x, 2) = (x^2 - 1)^2 + 2x^2 = x^4 + 1 \geq 1$, with equality at $x = 0$. Hence $\theta(2) = 1 = p^*$, and weak duality proves strong duality directly. Failure of sufficient hypotheses therefore does not decide the outcome.

Now keep both convexity and Slater, so that the theorem applies, and ask which optimum it guarantees to be attained.

Example 19.4 (Slater does not give primal attainment)**.** Consider

$$\inf_{x\in\mathbb{R}} e^{-x} \quad \text{subject to} \quad -x \le 0.$$

The data are convex, and $x = 1$ is strictly feasible. Feasible $x \to +\infty$ gives $p^* = 0$, but the exponential is positive at every finite point. At $\nu = 0$, $\theta(0) = \inf_x e^{-x} = 0$. For $\nu > 0$, $e^{-x} - \nu x \to -\infty$ as $x \to +\infty$, so the unique dual optimizer is $\nu^* = 0$. The theorem gives exactly what occurs: zero gap and dual attainment, without a primal optimizer.

Example 19.5 (Zero gap need not give a dual optimizer)**.** Recall the Chapter 18 problem $\min x$ subject to $x^2 \le 0$. Its primal optimizer is 0, while

$$\theta(0) = -\infty, \qquad \theta(\nu) = -\frac{1}{4\nu} \quad (\nu > 0), \qquad x(\nu) = -\frac{1}{2\nu}.$$

Thus $d^* = p^* = 0$, but no finite multiplier attains the dual supremum. The inner minimizers approach feasibility only as $\nu \to \infty$. Here convexity holds and Slater fails. Equality of values alone cannot supply an optimal primal–dual triple.

19.4 Linear programming as a special case

The examples show why convexity alone does not settle either gap or attainment. We next ask what the additional structure of a linear program allows us to prove. We will derive its dual explicitly, then use polyhedral closedness and the affine-constraint optimality result from Chapter 14 to obtain matching optimizers without requiring strict feasibility of every displayed inequality.

19.4.1 Deriving the dual and the bidual

Consider the standard-form linear program

$$(P) \qquad \inf_{\boldsymbol{x}\in\mathbb{R}^n} \boldsymbol{c}^{\mathsf{T}}\boldsymbol{x} \quad \text{subject to} \quad \boldsymbol{B}\boldsymbol{x} = \boldsymbol{b}, \quad \boldsymbol{x} \ge \boldsymbol{0}. \tag{19.4}$$

To keep every restriction explicit, write nonnegativity as $-\boldsymbol{x} \le \boldsymbol{0}$ and introduce a multiplier $\boldsymbol{\nu} \ge \boldsymbol{0}$. Its full-space Lagrangian is

$$L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = (\boldsymbol{c} + \boldsymbol{B}^{\mathsf{T}}\boldsymbol{\lambda} - \boldsymbol{\nu})^{\mathsf{T}}\boldsymbol{x} - \boldsymbol{b}^{\mathsf{T}}\boldsymbol{\lambda}.$$

The infimum of a nonconstant affine function on $\mathbb{R}^n$ is $-\infty$. A finite dual value therefore requires $\boldsymbol{\nu} = \boldsymbol{c} + \boldsymbol{B}^{\mathsf{T}}\boldsymbol{\lambda} \ge \boldsymbol{0}$, in which case the value is $-\boldsymbol{b}^{\mathsf{T}}\boldsymbol{\lambda}$. Eliminating $\boldsymbol{\nu}$ gives

$$(D) \qquad \sup_{\boldsymbol{\lambda}\in\mathbb{R}^m} -\boldsymbol{b}^{\mathsf{T}}\boldsymbol{\lambda} \quad \text{subject to} \quad \boldsymbol{c} + \boldsymbol{B}^{\mathsf{T}}\boldsymbol{\lambda} \ge \boldsymbol{0}. \tag{19.5}$$

For a feasible pair, subtracting the two objectives also gives the useful identity

$$\boldsymbol{c}^{\mathsf{T}}\boldsymbol{x} + \boldsymbol{b}^{\mathsf{T}}\boldsymbol{\lambda} = \boldsymbol{x}^{\mathsf{T}}(\boldsymbol{c} + \boldsymbol{B}^{\mathsf{T}}\boldsymbol{\lambda}) \ge 0. \tag{19.6}$$

Matching values are therefore equivalent to componentwise complementarity $x_i(c_i+(\boldsymbol{B}^{\mathsf{T}}\boldsymbol{\lambda})_i) = 0$.

We can now dualize (D) itself. Convert its maximum to $\inf_{\boldsymbol{\lambda}} \boldsymbol{b}^{\mathsf{T}}\boldsymbol{\lambda}$ subject to $-\boldsymbol{c}-\boldsymbol{B}^{\mathsf{T}}\boldsymbol{\lambda} \le \boldsymbol{0}$, and use $\boldsymbol{y} \ge \boldsymbol{0}$ as its inequality multiplier. The resulting Lagrangian is

$$\mathcal{L}_D(\boldsymbol{\lambda}, \boldsymbol{y}) = -\boldsymbol{c}^{\mathsf{T}}\boldsymbol{y} + (\boldsymbol{b} - \boldsymbol{B}\boldsymbol{y})^{\mathsf{T}}\boldsymbol{\lambda}.$$

Its infimum over the free variable $\boldsymbol{\lambda}$ is finite exactly when $\boldsymbol{B}\boldsymbol{y} = \boldsymbol{b}$. Its dual is thus $\sup\{-\boldsymbol{c}^{\mathsf{T}}\boldsymbol{y} : \boldsymbol{B}\boldsymbol{y} = \boldsymbol{b},\ \boldsymbol{y} \ge \boldsymbol{0}\}$. After reversing the objective sign, this is precisely (P). This calculation establishes the bidual form.

19.4.2 Why the optimal values agree

To supply the missing argument, we need both attainment and multiplier existence. The first follows from the closedness of finitely generated cones established in Lemma 12.7; the second follows from the affine-constraint KKT result in Theorem 14.2, whose proof uses Farkas' lemma. Neither step requires strict positivity of a feasible $\boldsymbol{x}$.

Theorem 19.2 (Linear-programming strong duality)**.** *If either (P) or (D) is feasible and has a finite optimal value, then both problems attain finite optimal values and $p^* = d^*$. If one is unbounded in its optimization direction, the other is infeasible. Both can be infeasible.*

Proof. Suppose first that (P) is feasible and $p^* \in \mathbb{R}$. Form the cone

$$K = \{(\boldsymbol{B}\boldsymbol{x}, \boldsymbol{c}^{\mathsf{T}}\boldsymbol{x} + s) : \boldsymbol{x} \ge \boldsymbol{0},\ s \ge 0\} \subseteq \mathbb{R}^{m+1}.$$

It is generated by the n vectors $(\boldsymbol{B}\boldsymbol{e}_i, c_i)$ and the vector $(\boldsymbol{0}, 1)$, so it is closed. Choose a feasible sequence with $\boldsymbol{c}^{\mathsf{T}}\boldsymbol{x}^{(k)} \to p^*$. Then $(\boldsymbol{b}, \boldsymbol{c}^{\mathsf{T}}\boldsymbol{x}^{(k)}) \in K$ converges to $(\boldsymbol{b}, p^*) \in K$. Thus there exist $\boldsymbol{x}^* \ge \boldsymbol{0}$ and $s \ge 0$ with

$$\boldsymbol{B}\boldsymbol{x}^* = \boldsymbol{b}, \qquad \boldsymbol{c}^{\mathsf{T}}\boldsymbol{x}^* + s = p^*.$$

By the definition of p^*, the feasible objective is at least p^*; therefore $s = 0$ and $\boldsymbol{x}^*$ attains the infimum.

Apply affine-constraint KKT necessity at this minimizer. There exist $\boldsymbol{\lambda}^* \in \mathbb{R}^m$ and $\boldsymbol{\nu}^* \ge \boldsymbol{0}$ such that

$$\boldsymbol{c} + \boldsymbol{B}^{\mathsf{T}}\boldsymbol{\lambda}^* - \boldsymbol{\nu}^* = \boldsymbol{0}, \qquad (\boldsymbol{x}^*)^{\mathsf{T}}\boldsymbol{\nu}^* = 0.$$

The first relation makes $\boldsymbol{\lambda}^*$ feasible for (D); the second and Equation (19.6) give $\boldsymbol{c}^{\mathsf{T}}\boldsymbol{x}^* = -\boldsymbol{b}^{\mathsf{T}}\boldsymbol{\lambda}^*$. Weak duality now proves optimality of both points and equality of their values.

For completeness, finite attainment also holds for a linear objective on any nonempty polyhedron with finite infimum: split each free variable into a difference of nonnegative variables and introduce nonnegative slack variables for inequalities. This converts it to the standard form just considered, without changing objective values, and a minimizing point maps back to the original variables. Suppose now that (D) is feasible and $d^* \in \mathbb{R}$. Apply this fact to its minimization form to obtain $\boldsymbol{\lambda}^*$. KKT necessity for that polyhedron provides a multiplier $\boldsymbol{x}^* \ge \boldsymbol{0}$ with

$$\boldsymbol{b} - \boldsymbol{B}\boldsymbol{x}^* = \boldsymbol{0}, \qquad (\boldsymbol{x}^*)^{\mathsf{T}}(\boldsymbol{c} + \boldsymbol{B}^{\mathsf{T}}\boldsymbol{\lambda}^*) = 0.$$

Hence $\boldsymbol{x}^*$ is primal feasible and again the two objective values match.

If (P) is unbounded below, any feasible point of (D) would give a finite lower bound, contradicting weak duality. The analogous argument applies if (D) is unbounded above. Finally, $B = 0$, $b = 1$, and $c = -1$ give an infeasible primal equation $0x = 1$ and an infeasible dual inequality $-1 \geq 0$. □

As a quick calculation, take $\boldsymbol{c} = (1, 2)^{\mathsf{T}}$, $\boldsymbol{B} = (1, 1)$, and $b = 1$. The primal minimum is attained at $(1, 0)$ with value 1. The dual conditions are $1 + \lambda \geq 0$ and $2 + \lambda \geq 0$, so maximizing $-\lambda$ gives $\lambda^* = -1$ and the same value. The zero components in Equation (19.6) show exactly how the two solutions match. The proof above explains why this agreement extends to every feasible, bounded linear program, including degenerate ones.

19.5 From matching values to the KKT conditions

19.5.1 Where equality must occur

To connect the value certificate to first-order conditions, let $\boldsymbol{x}$ be primal feasible and let $(\boldsymbol{\lambda}, \boldsymbol{\nu})$ be dual feasible with finite $\theta(\boldsymbol{\lambda}, \boldsymbol{\nu})$. Chapter 18's value-error decomposition Equation (18.9) measured the candidate gap relative to the two optimal values. Here we insert the Lagrangian instead, to identify which optimality conditions make that gap vanish:

$$f(\boldsymbol{x}) - \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \underbrace{-\boldsymbol{\nu}^{\mathsf{T}} \boldsymbol{h}(\boldsymbol{x})}_{\text{complementarity gap}} + \underbrace{L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) - \inf_{\boldsymbol{y} \in \mathbb{R}^n} L(\boldsymbol{y}, \boldsymbol{\lambda}, \boldsymbol{\nu})}_{\text{Lagrangian minimization gap}}. \tag{19.7}$$

The equality uses $\boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{0}$. Both terms on the right are nonnegative: the first by the signs of $\boldsymbol{\nu}$ and $\boldsymbol{h}(\boldsymbol{x})$, the second by the definition of the infimum. A zero gap therefore makes both terms zero. Moreover, each product $\nu_j h_j(\boldsymbol{x})$ is nonpositive, so zero sum means componentwise complementary slackness.

Proposition 19.3 (Matching values and KKT)**.** *For a primal–dual feasible triple, finite matching values $f(\boldsymbol{x}^*) = \theta(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ are equivalent to complementary slackness together with global minimization of $L(\cdot, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ by $\boldsymbol{x}^*$. They certify primal and dual optimality with zero gap. If the data are differentiable at $\boldsymbol{x}^*$, they also imply KKT, without convexity or a constraint qualification. Conversely, in a differentiable convex program with affine equalities, any KKT triple has matching values.*

Proof. The equivalence follows from Equation (19.7). When the endpoint values match, Equation (19.1) forces equality throughout, proving optimality. Global minimization of a differentiable function on the full space gives

$$\nabla_{\boldsymbol{x}} L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = \nabla f(\boldsymbol{x}^*) + \sum_i \lambda_i^* \nabla g_i(\boldsymbol{x}^*) + \sum_j \nu_j^* \nabla h_j(\boldsymbol{x}^*) = \boldsymbol{0}. \tag{19.8}$$

Together with feasibility and complementary slackness this is KKT.

For the converse, fix the multipliers of a convex KKT triple. The function $L(\cdot, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ is convex because f, h_j are convex, $\boldsymbol{\nu}^* \geq \boldsymbol{0}$, and the equality terms are affine. For every $\boldsymbol{x}$, the first-order convexity inequality and stationarity give

$$L(\boldsymbol{x}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) \geq L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) + \nabla_{\boldsymbol{x}} L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)^{\mathsf{T}} (\boldsymbol{x} - \boldsymbol{x}^*) = L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*).$$

Thus $\boldsymbol{x}^*$ globally minimizes the Lagrangian. Complementary slackness then gives $\theta(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = f(\boldsymbol{x}^*)$. □

Convexity is needed only for this last upgrade from stationarity to global Lagrangian minimization. The KKT triple $(0, 0, 0)$ in Example 19.1 demonstrates why the upgrade fails in general. Conversely, strong duality by itself is a statement about values: to infer KKT from optimal points, both optima must be attained.

Corollary 19.4 (Existence of convex KKT multipliers under Slater)**.** *Under Theorem 19.1, suppose the primal infimum is attained at $\boldsymbol{x}^*$ and the data are differentiable there. Then some dual-optimal $(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ makes $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ a KKT triple.*

Proof. Slater's theorem supplies a dual optimizer with $\theta(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = d^* = p^* = f(\boldsymbol{x}^*)$. Apply Proposition 19.3. □

19.5.2 Selecting a primal point from the inner minimizers

Given dual-feasible multipliers, define

$$X(\widehat{\boldsymbol{\lambda}}, \widehat{\boldsymbol{\nu}}) = \arg\min_{\boldsymbol{x} \in \mathbb{R}^n} L(\boldsymbol{x}, \widehat{\boldsymbol{\lambda}}, \widehat{\boldsymbol{\nu}}).$$

The recovery criterion from Chapter 18 asks for a member of this set that is primal feasible and complementary. The preceding proposition explains why these checks are exactly the ones needed for a matching certificate. An arbitrary stationary point of a nonconvex Lagrangian does not suffice: its value need not be θ, even if primal feasibility holds. The following affine example isolates the selection issue without any nonlinear inner calculation.

Example 19.6 (An optimal multiplier is not yet a primal answer)**.** Consider $\min_{x \in \mathbb{R}} x$ subject to $1 - x \le 0$. The primal optimizer is $x^* = 1$ with $p^* = 1$, and

$$L(x, \nu) = (1 - \nu)x + \nu, \qquad \theta(\nu) = \begin{cases} 1, & \nu = 1, \\ -\infty, & \nu \neq 1. \end{cases}$$

The dual optimizer is $\nu^* = 1$, but $X(1) = \mathbb{R}$ because $L(x, 1) = 1$ for every x. The response $x = 0$ is infeasible. The response $x = 2$ is feasible but nonoptimal: it violates complementary slackness since $\nu^*(1 - x) = -1$. Only $x = 1$ passes both checks.

If both optima are attained with zero gap, every primal optimizer belongs to $X(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$, by the matching-value argument. Therefore a unique member of that set must be the primal optimizer. Without uniqueness, primal recovery requires selection; neither exact dual optimization nor an arbitrary exact inner solve performs that selection automatically.

19.5.3 An approximate stopping certificate

For computations, we can reuse the certificate already derived in Equation (18.9). With a feasible candidate pair and finite $G = f(\widehat{\boldsymbol{x}}) - \theta(\widehat{\boldsymbol{\lambda}}, \widehat{\boldsymbol{\nu}})$, its three nonnegative parts satisfy

$$0 \le f(\widehat{\boldsymbol{x}}) - p^* \le G, \tag{19.9}$$

$$0 \le d^* - \theta(\widehat{\boldsymbol{\lambda}}, \widehat{\boldsymbol{\nu}}) \le G, \tag{19.10}$$

$$0 \le p^* - d^* \le G. \tag{19.11}$$

Thus the stopping test

$$G \le \varepsilon_{\text{out}} \tag{19.12}$$

certifies all three value errors, independently of convexity. Its use here requires the same distinction as in Chapter 18: an inexact inner objective is an upper bound on θ, not automatically a lower bound on p^*. When an inner objective-error bound is available, use the corrected lower bound in Equation (18.21). Later in this chapter we will examine how inner error also affects the ALM multiplier step itself.

19.6 Global saddle points

19.6.1 From attained zero gap to the saddle inequalities

Assume both optima are attained and $p^* = d^* \in \mathbb{R}$. Fix a primal optimizer $\boldsymbol{x}^*$ and a dual optimizer $(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$. The matching-value argument gives

$$L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = f(\boldsymbol{x}^*) = \theta(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*).$$

For any dual-feasible $(\boldsymbol{\lambda}, \boldsymbol{\nu})$, feasibility of $\boldsymbol{x}^*$ gives $L(\boldsymbol{x}^*, \boldsymbol{\lambda}, \boldsymbol{\nu}) \le f(\boldsymbol{x}^*)$. For any $\boldsymbol{x} \in \mathbb{R}^n$, the definition of the dual infimum gives $\theta(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) \le L(\boldsymbol{x}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$. Combining the two comparisons produces

$$L(\boldsymbol{x}^*, \boldsymbol{\lambda}, \boldsymbol{\nu}) \le L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) \le L(\boldsymbol{x}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) \tag{19.13}$$

for every $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{\lambda} \in \mathbb{R}^m$, and $\boldsymbol{\nu} \ge \mathbf{0}$. A triple with $\boldsymbol{\nu}^* \ge \mathbf{0}$ satisfying these inequalities is called a *global Lagrangian saddle point*: it minimizes in the primal variables and maximizes in the multipliers when the other block is fixed. The word global refers to the quantifiers over all points, not just nearby ones.

Theorem 19.5 (Saddle points and attained zero gap)**.** *A triple is a global Lagrangian saddle point if and only if it consists of primal and dual optimizers with a common finite optimal value.*

Proof. The preceding derivation proves the implication from attained zero gap. For the converse, suppose Equation (19.13) holds with $\boldsymbol{\nu}^* \ge \mathbf{0}$. If $g_i(\boldsymbol{x}^*) \ne 0$, varying the unrestricted λ_i makes the left expression arbitrarily large, a contradiction. If $h_j(\boldsymbol{x}^*) > 0$, sending $\nu_j \to +\infty$ gives the same contradiction. Thus $\boldsymbol{x}^*$ is primal feasible. Set $\boldsymbol{\lambda} = \boldsymbol{\lambda}^*$ and $\boldsymbol{\nu} = \mathbf{0}$ in the left inequality. Feasibility then gives

$$f(\boldsymbol{x}^*) \le f(\boldsymbol{x}^*) + (\boldsymbol{\nu}^*)^{\mathsf{T}} \boldsymbol{h}(\boldsymbol{x}^*) \le f(\boldsymbol{x}^*).$$

Hence complementary slackness holds and the middle value equals $f(\boldsymbol{x}^*)$. The right inequality states that $\boldsymbol{x}^*$ globally minimizes the Lagrangian, so its value is $\theta(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$. We have obtained finite matching values; Equation (19.1) proves both optimality statements. □

Corollary 19.6 (KKT and saddle points)**.** *For differentiable data, every global Lagrangian saddle point satisfies KKT. In a convex program with affine equalities, the converse also holds.*

Proof. The saddle theorem gives finite matching values, so Proposition 19.3 gives KKT. Under the convex assumptions, the same proposition gives matching values from KKT, and the saddle theorem then supplies the global saddle inequalities. □

We can also ask what remains true if only the primal comparison is local. Suppose the left inequality in Equation (19.13) holds for every sign-admissible multiplier, while the right inequality is required only for $\boldsymbol{x}$ in a neighbourhood of $\boldsymbol{x}^*$. The left inequality still forces primal feasibility and complementary slackness by the same multiplier variations as above. The right inequality now gives a local minimum of $L(\cdot, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$. If the data are differentiable at $\boldsymbol{x}^*$, its gradient vanishes, and KKT follows. This local statement does not provide global Lagrangian minimization or a matching-value certificate. The differentiability condition is needed for this classical KKT conclusion.

This combines two different kinds of reasoning. First-order conditions refer to derivatives at one point; the saddle inequalities compare values globally. Convexity connects them. Slater is a condition ensuring the existence of the needed multipliers at an attained primal optimum, not an extra hypothesis for the sufficiency of an already available convex KKT triple.

19.6.2 A landscape and the full Lagrangian Hessian

To visualize the saddle, consider

$$\min_{x\in\mathbb{R}} \frac{1}{2}x^2 \quad \text{subject to} \quad x = 1.$$

Completing the square gives

$$L(x,\lambda) = \frac{1}{2}x^2 + \lambda(x-1) = \frac{1}{2}(x+\lambda)^2 - \frac{1}{2}\lambda^2 - \lambda, \qquad \theta(\lambda) = \frac{1}{2} - \frac{1}{2}(\lambda+1)^2.$$

Thus $x(\lambda) = -\lambda$, and the dual maximum at $\lambda^* = -1$ matches $f(1) = 1/2$. At the optimal coordinates,

$$L(1,\lambda) = \frac{1}{2}, \qquad L(x,-1) = \frac{1}{2} + \frac{1}{2}(x-1)^2 \geq \frac{1}{2}.$$

The multiplier section is constant, whereas the primal section is minimized at $x = 1$, exactly as the saddle inequalities require.

To connect this picture to LN, distinguish the primal Hessian $\boldsymbol{H}_L = \nabla^2_{xx} L = 1$ from the Hessian in both variables:

$$\nabla^2_{(x,\lambda)} L = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma(\nabla^2_{(x,\lambda)} L) = \left\{ \frac{1+\sqrt{5}}{2}, \frac{1-\sqrt{5}}{2} \right\}.$$

The full Hessian is indefinite; see Figure 19.2 for an illustration. More generally, for twice differentiable equality-constrained data, put $\boldsymbol{A}(\boldsymbol{x}) = [\nabla g_1(\boldsymbol{x}), \ldots, \nabla g_m(\boldsymbol{x})]$ as in LN. The Newton matrix for the KKT equations is

$$\nabla^2_{(\boldsymbol{x},\boldsymbol{\lambda})} L = \begin{pmatrix} \boldsymbol{H}_L & \boldsymbol{A} \\ \boldsymbol{A}^\mathsf{T} & 0 \end{pmatrix}.$$

Whenever $\boldsymbol{A} \neq 0$, choose $\boldsymbol{u}, \boldsymbol{v}$ with $\boldsymbol{u}^\mathsf{T}\boldsymbol{A}\boldsymbol{v} \neq 0$. The quadratic form at $(\boldsymbol{u}, t\boldsymbol{v})$ is $\boldsymbol{u}^\mathsf{T}\boldsymbol{H}_L\boldsymbol{u} + 2t\boldsymbol{u}^\mathsf{T}\boldsymbol{A}\boldsymbol{v}$, which takes both signs as t varies. The matrix is therefore indefinite even when $\boldsymbol{H}_L$ is positive definite. A KKT point need not minimize L in the combined variables, explaining why a line search requiring joint Lagrangian decrease is not an appropriate stabilization of LN.

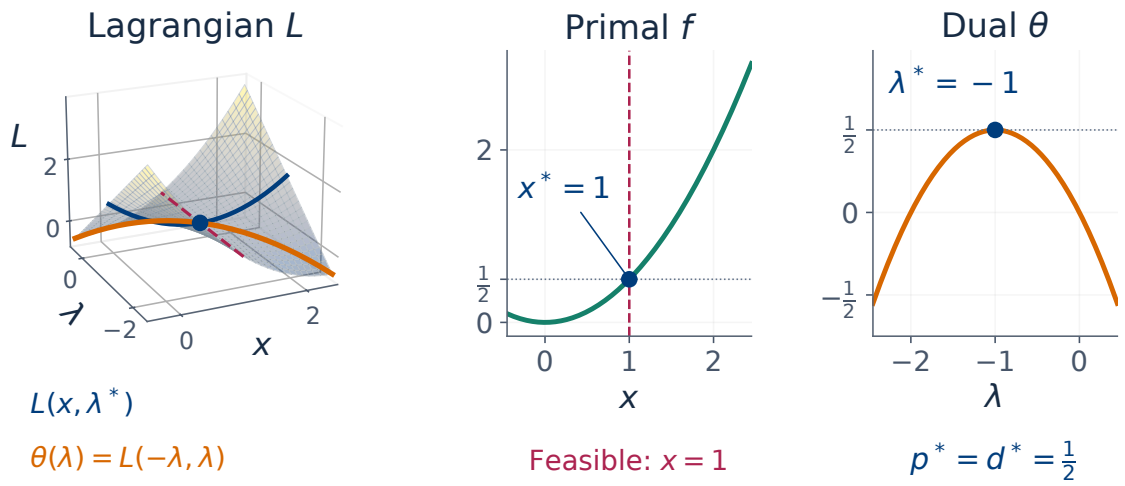


Figure 19.2: The Lagrangian for $\min \frac{1}{2}x^2$ subject to $x = 1$. The blue section is $L(x, \lambda^*)$ at $\lambda^* = -1$; the orange minimizing-response curve has coordinates $(x, \lambda) = (-\lambda, \lambda)$ and height $\theta(\lambda)$; the red line fixes $x = 1$ and has constant height $L(1, \lambda) = 1/2$. They meet at the saddle pair $(1, -1)$. These curves show the primal minimum at the optimal multiplier and the dual values obtained by minimizing over x.

19.7 From saddle geometry to algorithms

19.7.1 Why descent in one block and ascent in the other can fail

The saddle inequalities suggest decreasing L in $\boldsymbol{x}$ and increasing it in the multipliers. Before treating this as an algorithmic guarantee, consider the simplest equality problem

$$\min_{x \in \mathbb{R}} 0 \quad \text{subject to} \quad x = 0, \qquad L(x, \lambda) = x\lambda.$$

It has the unique saddle pair $(0, 0)$ and zero gap. A simultaneous gradient step with fixed $\eta > 0$ is

$$x^{(k+1)} = x^{(k)} - \eta\lambda^{(k)}, \qquad \lambda^{(k+1)} = \lambda^{(k)} + \eta x^{(k)}.$$

Squaring and adding cancels the cross terms:

$$(x^{(k+1)})^2 + (\lambda^{(k+1)})^2 = (1 + \eta^2)((x^{(k)})^2 + (\lambda^{(k)})^2).$$

Thus every nonzero start diverges, even though the problem is convex and the saddle exists.

One might instead update the multiplier first and use it in the primal step:

$$\lambda^{(k+1)} = \lambda^{(k)} + \eta x^{(k)}, \qquad x^{(k+1)} = x^{(k)} - \eta\lambda^{(k+1)}.$$

The update matrix is $\begin{pmatrix} 1-\eta^2 & -\eta \\ \eta & 1 \end{pmatrix}$. Its determinant is 1, and for $0 < \eta < 2$ its two eigenvalues are distinct complex conjugates of modulus 1. More explicitly, direct substitution shows that the positive definite quadratic expression

$$E(x, \lambda) = x^2 + \lambda^2 + \eta x\lambda$$

is preserved when $0 < \eta < 2$. Every nonzero trajectory therefore stays bounded but fails to approach $(0, 0)$. At $\eta = 1/2$ it circulates on an ellipse; it need not be exactly periodic. At $\eta = 1$, the update matrix satisfies $M^6 = I$, producing a six-step cycle from every nonzero start. Updating one block after the other has removed divergence, but has not created convergence.

19.7.2 How the augmented term changes this example

For the same problem, ALM with $\rho > 0$ exactly minimizes

$$L_A(x, \lambda^{(k)}; \rho) = \lambda^{(k)} x + \frac{\rho}{2} x^2.$$

Its minimizer and multiplier update are

$$x^{(k+1)} = -\frac{\lambda^{(k)}}{\rho}, \qquad \lambda^{(k+1)} = \lambda^{(k)} + \rho x^{(k+1)} = 0.$$

The next primal minimization returns $x^{(k+2)} = 0$. The added curvature makes the primal response well defined and the residual correction removes the multiplier error. Figure 19.3 compares these three iterations for the same starting pair.

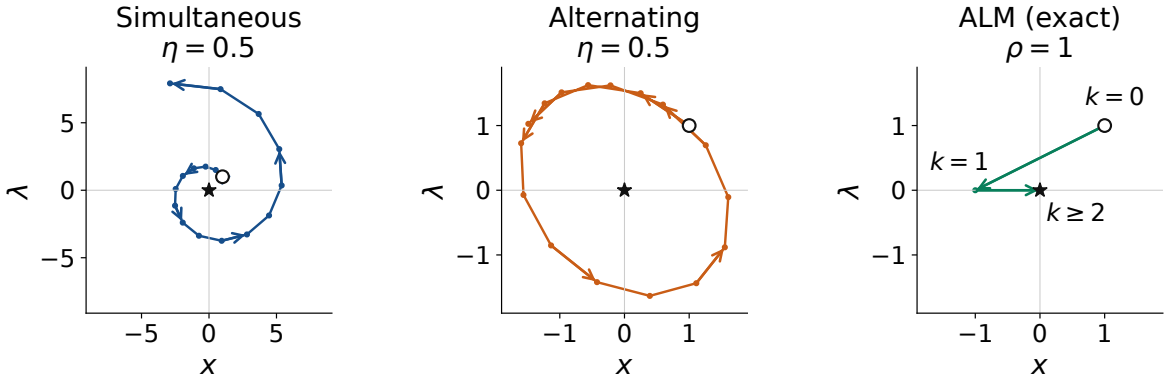


Figure 19.3: Primal–dual trajectories for $L(x, \lambda) = x\lambda$. Simultaneous descent–ascent diverges, alternating updates remain on a nonzero closed invariant curve, and exact ALM reaches the saddle after the multiplier correction and the following primal minimization. The gradient step length is $\eta = 1/2$ and the ALM penalty is $\rho = 1$.

19.7.3 From the ALM subproblem to a regularized saddle point

We now derive the mechanism behind the preceding example. For this part, let $f : \mathbb{R}^n \to \mathbb{R}$ be convex and differentiable, and let the equality residual be $\boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{B}\boldsymbol{x} - \boldsymbol{b}$. Fix $\rho > 0$ and a current multiplier $\boldsymbol{\lambda}^{(k)}$. As in Chapter 15, assume that the augmented subproblem has an attained global minimum and perform

$$\boldsymbol{x}^{(k+1)} \in \arg\min_{\boldsymbol{x}} L_A(\boldsymbol{x}, \boldsymbol{\lambda}^{(k)}; \rho), \qquad \boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho \boldsymbol{g}(\boldsymbol{x}^{(k+1)}),$$

where

$$L_A(\boldsymbol{x}, \boldsymbol{\lambda}; \rho) = f(\boldsymbol{x}) + \boldsymbol{\lambda}^{\mathsf{T}} \boldsymbol{g}(\boldsymbol{x}) + \frac{\rho}{2} \|\boldsymbol{g}(\boldsymbol{x})\|^2.$$

An attained exact inner solve is an assumption about this subproblem; it must not be inferred merely from strong duality of the original problem. For the next calculations, abbreviate the computed pair by $\boldsymbol{x}^+ = \boldsymbol{x}^{(k+1)}$ and $\boldsymbol{\lambda}^+ = \boldsymbol{\lambda}^{(k+1)}$.

To relate these updates to saddle geometry, keep the old multiplier as a center and introduce the two-variable function

$$\Phi_k(\boldsymbol{x}, \boldsymbol{z}) = L(\boldsymbol{x}, \boldsymbol{z}) - \frac{1}{2\rho} \left\| \boldsymbol{z} - \boldsymbol{\lambda}^{(k)} \right\|^2. \tag{19.14}$$

For fixed $\boldsymbol{x}$, complete the square in $\boldsymbol{z}$:

$$\Phi_k(\boldsymbol{x}, \boldsymbol{z}) = L_A(\boldsymbol{x}, \boldsymbol{\lambda}^{(k)}; \rho) - \frac{1}{2\rho} \left\| \boldsymbol{z} - (\boldsymbol{\lambda}^{(k)} + \rho \boldsymbol{g}(\boldsymbol{x})) \right\|^2. \tag{19.15}$$

Its unique maximizing multiplier is consequently $\boldsymbol{\lambda}^{(k)} + \rho \boldsymbol{g}(\boldsymbol{x})$, and its maximum value is $L_A(\boldsymbol{x}, \boldsymbol{\lambda}^{(k)}; \rho)$. At $\boldsymbol{x} = \boldsymbol{x}^+$, the maximizing multiplier is exactly the ALM correction $\boldsymbol{\lambda}^+$.

It remains to check the other block. Stationarity of the solved augmented subproblem gives

$$\mathbf{0} = \nabla f(\boldsymbol{x}^+) + \boldsymbol{B}^\mathsf{T}(\boldsymbol{\lambda}^{(k)} + \rho \boldsymbol{g}(\boldsymbol{x}^+)) = \nabla_{\boldsymbol{x}} L(\boldsymbol{x}^+, \boldsymbol{\lambda}^+).$$

Because $L(\cdot, \boldsymbol{\lambda}^+)$ is convex, $\boldsymbol{x}^+$ globally minimizes this ordinary Lagrangian. Subtracting the constant proximal term does not change its minimizers. We have therefore verified both inequalities

$$\Phi_k(\boldsymbol{x}^+, \boldsymbol{z}) \le \Phi_k(\boldsymbol{x}^+, \boldsymbol{\lambda}^+) \le \Phi_k(\boldsymbol{x}, \boldsymbol{\lambda}^+) \qquad (\boldsymbol{x} \in \mathbb{R}^n,\ \boldsymbol{z} \in \mathbb{R}^m). \tag{19.16}$$

Thus one exact ALM step finds a global saddle point of Φ_k. This is a regularized saddle problem with a center depending on k; the computed $\boldsymbol{x}^+$ need not yet satisfy the original equality.

We can now take the two orders of optimization without an unjustified interchange. Each is bounded on both sides by the middle saddle value in Equation (19.16), so

$$\begin{aligned} \min_{\boldsymbol{x}} L_A(\boldsymbol{x}, \boldsymbol{\lambda}^{(k)}; \rho) &= \inf_{\boldsymbol{x}} \sup_{\boldsymbol{z}} \Phi_k(\boldsymbol{x}, \boldsymbol{z}) \\ &= \Phi_k(\boldsymbol{x}^+, \boldsymbol{\lambda}^+) \\ &= \sup_{\boldsymbol{z}} \inf_{\boldsymbol{x}} \Phi_k(\boldsymbol{x}, \boldsymbol{z}) \\ &= \max_{\boldsymbol{z}} \left\{ \theta(\boldsymbol{z}) - \frac{1}{2\rho} \left\| \boldsymbol{z} - \boldsymbol{\lambda}^{(k)} \right\|^2 \right\}. \end{aligned}$$

The first equality uses the completed square; the last uses that the quadratic term is independent of $\boldsymbol{x}$. The saddle point proves both middle equality and attainment of the last maximum. In particular,

$$\boldsymbol{\lambda}^+ = \arg\max_{\boldsymbol{z} \in \mathbb{R}^m} \left\{ \theta(\boldsymbol{z}) - \frac{1}{2\rho} \left\| \boldsymbol{z} - \boldsymbol{\lambda}^{(k)} \right\|^2 \right\}. \tag{19.17}$$

This is a *proximal step* for maximizing the ordinary dual function: the quadratic term penalizes movement from the current multiplier. The primal computation in ALM evaluates this dual step without requiring an explicit formula for θ. No primal or dual optimizer of the original problem was needed for this single-step identity.

19.7.4 How the proximal term damps the multiplier update

To quantify the effect of this regularization, put $\boldsymbol{r} = \boldsymbol{g}(\boldsymbol{x}^+)$. Since $\boldsymbol{x}^+$ minimizes the ordinary Lagrangian at $\boldsymbol{\lambda}^+$, evaluating another multiplier's inner infimum at this same point gives

$$\theta(\boldsymbol{z}) \le L(\boldsymbol{x}^+, \boldsymbol{z}) = \theta(\boldsymbol{\lambda}^+) + \boldsymbol{r}^\mathsf{T}(\boldsymbol{z} - \boldsymbol{\lambda}^+). \tag{19.18}$$

Hence $\boldsymbol{r}$ is a supergradient at the *new* multiplier. Substitute $\boldsymbol{r} = (\boldsymbol{\lambda}^+ - \boldsymbol{\lambda}^{(k)})/\rho$ and complete the square to obtain

$$\theta(\boldsymbol{z}) - \frac{1}{2\rho} \left\| \boldsymbol{z} - \boldsymbol{\lambda}^{(k)} \right\|^2 \le \theta(\boldsymbol{\lambda}^+) - \frac{1}{2\rho} \left\| \boldsymbol{\lambda}^+ - \boldsymbol{\lambda}^{(k)} \right\|^2 - \frac{1}{2\rho} \left\| \boldsymbol{z} - \boldsymbol{\lambda}^+ \right\|^2.$$

The last term proves uniqueness of the proximal maximizer and measures the loss incurred by choosing another multiplier. The update can equivalently be written

$$\frac{\boldsymbol{\lambda}^{(k+1)} - \boldsymbol{\lambda}^{(k)}}{\rho} \in \partial^{+}\theta(\boldsymbol{\lambda}^{(k+1)}).$$

If θ is differentiable there, this becomes

$$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho \nabla\theta(\boldsymbol{\lambda}^{(k+1)}).$$

It is an implicit ascent step: the new multiplier, rather than the old one, is used to evaluate the first-order information. Thus damping here means the stability created by solving this implicit relation; it does not mean that an arbitrary explicit ascent step has simply been multiplied by a smaller scalar.

For a stability estimate relative to the solution, now additionally assume that a finite dual optimum is attained at $\boldsymbol{\lambda}^{*}$. In the present convex setting, an existing primal KKT pair is one sufficient way to obtain such a multiplier. Set $\boldsymbol{z} = \boldsymbol{\lambda}^{*}$ in Equation (19.18). Since $\theta(\boldsymbol{\lambda}^{*}) \geq \theta(\boldsymbol{\lambda}^{+})$, it follows that

$$(\boldsymbol{\lambda}^{+} - \boldsymbol{\lambda}^{(k)})^{\mathsf{T}}(\boldsymbol{\lambda}^{+} - \boldsymbol{\lambda}^{*}) \leq 0.$$

Expand the squared distance from the old iterate to $\boldsymbol{\lambda}^{*}$:

$$\left\|\boldsymbol{\lambda}^{(k)} - \boldsymbol{\lambda}^{*}\right\|^{2} = \left\|\boldsymbol{\lambda}^{+} - \boldsymbol{\lambda}^{*}\right\|^{2} + \left\|\boldsymbol{\lambda}^{+} - \boldsymbol{\lambda}^{(k)}\right\|^{2} - 2(\boldsymbol{\lambda}^{+} - \boldsymbol{\lambda}^{(k)})^{\mathsf{T}}(\boldsymbol{\lambda}^{+} - \boldsymbol{\lambda}^{*}).$$

The cross term has the required sign, giving the Fejér inequality

$$\left\|\boldsymbol{\lambda}^{(k+1)} - \boldsymbol{\lambda}^{*}\right\|^{2} + \left\|\boldsymbol{\lambda}^{(k+1)} - \boldsymbol{\lambda}^{(k)}\right\|^{2} \leq \left\|\boldsymbol{\lambda}^{(k)} - \boldsymbol{\lambda}^{*}\right\|^{2}. \tag{19.19}$$

A nonzero update strictly reduces distance to every dual optimizer, which rules out the persistent cycles seen in the alternating gradient example. Summing over k bounds the sum of squared multiplier increments. With fixed $\rho > 0$, the ALM correction therefore gives

$$\boldsymbol{g}(\boldsymbol{x}^{(k+1)}) = \frac{\boldsymbol{\lambda}^{(k+1)} - \boldsymbol{\lambda}^{(k)}}{\rho} \longrightarrow \mathbf{0}.$$

No increase of ρ was needed. A proof that the multipliers themselves converge will follow from continuity of the proximal map below; primal iterate convergence still requires additional properties of the inner minimizers.

The common quadratic test of Chapter 15 makes the parameter's effect explicit. For

$$f(\boldsymbol{x}) = \frac{1}{2}\left[(x_1 - 2)^2 + 4(x_2 - 1)^2\right], \qquad g(\boldsymbol{x}) = x_1 + x_2 - 1,$$

ordinary Lagrangian stationarity at the corrected multiplier gives $x_1^{(k+1)} = 2 - \lambda^{(k+1)}$ and $x_2^{(k+1)} = 1 - \lambda^{(k+1)}/4$. Substitution into the update yields

$$\lambda^{(k+1)} = \lambda^{(k)} + \rho\left(2 - \frac{5}{4}\lambda^{(k+1)}\right).$$

Since $\lambda^{*} = 8/5$, subtraction gives

$$e^{(k+1)} = \frac{e^{(k)}}{1 + 5\rho/4}, \qquad e^{(k)} = \lambda^{(k)} - \lambda^{*}. \tag{19.20}$$

Every fixed positive ρ gives a contraction in this example. Increasing ρ improves this particular outer contraction factor while *weakening* the proximal penalty $1/(2\rho)$. It simultaneously changes the inner Hessian to

$$\nabla^2_{\boldsymbol{xx}} L_A = \nabla^2 f + \rho \boldsymbol{B}^\mathsf{T} \boldsymbol{B},$$

which can become poorly conditioned, as quantified in Chapter 15. Stable outer progress does not remove the cost of the inner solve.

19.7.5 Why optimizing the regularized dual produces a smooth function

There is a second interpretation of the same step. Instead of recording the maximizing point in the proximal problem, record its optimized value:

$$\begin{aligned} \theta_\rho(\boldsymbol{\lambda}) &= \max_{\boldsymbol{z}\in\mathbb{R}^m} \left\{ \theta(\boldsymbol{z}) - \frac{1}{2\rho} \|\boldsymbol{z} - \boldsymbol{\lambda}\|^2 \right\}, \\ T_\rho(\boldsymbol{\lambda}) &= \arg\max_{\boldsymbol{z}\in\mathbb{R}^m} \left\{ \theta(\boldsymbol{z}) - \frac{1}{2\rho} \|\boldsymbol{z} - \boldsymbol{\lambda}\|^2 \right\}. \end{aligned} \tag{19.21}$$

The value θ_ρ is the concave Moreau envelope, equivalently the negative of the usual convex Moreau envelope of $-\theta$. We will derive its smoothness explicitly, rather than assuming that subtracting a quadratic has made the inner function smooth. The latter claim would be false: a kink in $\theta(\boldsymbol{z})$ remains a kink after subtracting the quadratic.

First separate this construction from its ALM implementation. In the next lemma, $\theta : \mathbb{R}^m \to \mathbb{R} \cup \{-\infty\}$ is any *proper closed concave function*: it is not identically $-\infty$ and is upper semicontinuous. The ordinary dual has closedness and concavity by Theorem 18.3 and proposition 18.4. It is proper whenever one finite Lagrangian lower bound exists; in particular, the exact ALM step above supplies one at $\boldsymbol{\lambda}^+$. Neither attainment of the original primal optimum nor differentiability of θ is required for the envelope construction.

Lemma 19.7 (Proximal map and smooth dual envelope)**.** *For a proper closed concave θ and $\rho > 0$, the maximum in Equation (19.21) is finite and uniquely attained for every $\boldsymbol{\lambda} \in \mathbb{R}^m$. Its maximizer satisfies*

$$\|T_\rho(\boldsymbol{\lambda}) - T_\rho(\boldsymbol{\mu})\|^2 \le (T_\rho(\boldsymbol{\lambda}) - T_\rho(\boldsymbol{\mu}))^\mathsf{T} (\boldsymbol{\lambda} - \boldsymbol{\mu}). \tag{19.22}$$

The envelope is concave and continuously differentiable, with

$$\begin{aligned} \nabla\theta_\rho(\boldsymbol{\lambda}) &= \frac{T_\rho(\boldsymbol{\lambda}) - \boldsymbol{\lambda}}{\rho}, \\ \|\nabla\theta_\rho(\boldsymbol{\lambda}) - \nabla\theta_\rho(\boldsymbol{\mu})\| &\le \frac{1}{\rho} \|\boldsymbol{\lambda} - \boldsymbol{\mu}\|. \end{aligned} \tag{19.23}$$

Proof. A proper closed convex function has an affine minorant, obtained by separating its epigraph from a point strictly below it. Applied to $-\theta$, this gives an affine majorant of θ. The negative quadratic in Equation (19.21) then forces the maximized expression to tend to $-\infty$ as $\|\boldsymbol{z}\| \to \infty$. Upper semicontinuity and a point with finite θ give a finite attained maximum. Strict concavity on its convex effective domain makes the maximizer unique. Write $\boldsymbol{p} = T_\rho(\boldsymbol{\lambda})$.

We next derive the first-order information at $\boldsymbol{p}$ without assuming differentiability. For any $\boldsymbol{z}$ with finite $\theta(\boldsymbol{z})$ and $0 < t < 1$, test the proximal objective at $\boldsymbol{p} + t(\boldsymbol{z} - \boldsymbol{p})$. Optimality of

$\boldsymbol{p}$ and concavity of θ give, after expanding the quadratic and dividing by t,

$$\theta(\boldsymbol{z}) - \theta(\boldsymbol{p}) \le \frac{1}{\rho}(\boldsymbol{p} - \boldsymbol{\lambda})^{\mathsf{T}}(\boldsymbol{z} - \boldsymbol{p}) + \frac{t}{2\rho}\left\|\boldsymbol{z} - \boldsymbol{p}\right\|^2.$$

Letting $t \downarrow 0$ proves that $(\boldsymbol{p} - \boldsymbol{\lambda})/\rho \in \partial^+\theta(\boldsymbol{p})$. For $\boldsymbol{q} = T_\rho(\boldsymbol{\mu})$, apply the two supergradient inequalities between $\boldsymbol{p}$ and $\boldsymbol{q}$ and add them. The result is

$$\big((\boldsymbol{p} - \boldsymbol{\lambda}) - (\boldsymbol{q} - \boldsymbol{\mu})\big)^{\mathsf{T}}(\boldsymbol{p} - \boldsymbol{q}) \le 0,$$

which rearranges to Equation (19.22). Cauchy–Schwarz implies $\|\boldsymbol{p} - \boldsymbol{q}\| \le \|\boldsymbol{\lambda} - \boldsymbol{\mu}\|$, so T_ρ is continuous. Expanding another square also gives

$$\begin{aligned}\left\|(\boldsymbol{p} - \boldsymbol{\lambda}) - (\boldsymbol{q} - \boldsymbol{\mu})\right\|^2 &\le \left\|\boldsymbol{\lambda} - \boldsymbol{\mu}\right\|^2 - \left\|\boldsymbol{p} - \boldsymbol{q}\right\|^2 \\ &\le \left\|\boldsymbol{\lambda} - \boldsymbol{\mu}\right\|^2.\end{aligned}$$

To differentiate the optimized value, set $\boldsymbol{p}' = T_\rho(\boldsymbol{\lambda} + \boldsymbol{v})$. Testing the maximum at $\boldsymbol{p}$ for the new center, and at $\boldsymbol{p}'$ for the old center, respectively, gives

$$\begin{aligned}\theta_\rho(\boldsymbol{\lambda} + \boldsymbol{v}) - \theta_\rho(\boldsymbol{\lambda}) &\ge \frac{1}{\rho}(\boldsymbol{p} - \boldsymbol{\lambda})^{\mathsf{T}}\boldsymbol{v} - \frac{1}{2\rho}\left\|\boldsymbol{v}\right\|^2, \\ \theta_\rho(\boldsymbol{\lambda} + \boldsymbol{v}) - \theta_\rho(\boldsymbol{\lambda}) &\le \frac{1}{\rho}(\boldsymbol{p}' - \boldsymbol{\lambda})^{\mathsf{T}}\boldsymbol{v} - \frac{1}{2\rho}\left\|\boldsymbol{v}\right\|^2.\end{aligned}$$

Since $\|\boldsymbol{p}' - \boldsymbol{p}\| \le \|\boldsymbol{v}\|$, these inequalities place the remainder after subtracting $(\boldsymbol{p} - \boldsymbol{\lambda})^{\mathsf{T}}\boldsymbol{v}/\rho$ between $-\left\|\boldsymbol{v}\right\|^2/(2\rho)$ and $\left\|\boldsymbol{v}\right\|^2/(2\rho)$. This proves the gradient formula. The preceding square estimate, divided by ρ^2, proves its Lipschitz bound and hence continuity. Finally, the expression maximized in Equation (19.21) is jointly concave in $(\boldsymbol{z}, \boldsymbol{\lambda})$. Testing it at the convex combination of two maximizing pairs proves concavity of the optimized value θ_ρ. □

We can identify this envelope with the augmented dual at every center, even when a particular augmented infimum is not attained. Introduce a separate residual variable $\boldsymbol{u}$ and consider

$$\inf_{\boldsymbol{x},\boldsymbol{u}}\left\{f(\boldsymbol{x}) + \boldsymbol{\lambda}^{\mathsf{T}}\boldsymbol{u} + \frac{\rho}{2}\left\|\boldsymbol{u}\right\|^2 : \boldsymbol{B}\boldsymbol{x} - \boldsymbol{b} - \boldsymbol{u} = \boldsymbol{0}\right\}.$$

This has a finite convex objective on the full space and a feasible affine equality. Its infimum is finite: if $\theta(\boldsymbol{z}_0) > -\infty$, then completing the residual square gives the lower bound

$$L_A(\boldsymbol{x}, \boldsymbol{\lambda}; \rho) \ge \theta(\boldsymbol{z}_0) - \frac{1}{2\rho}\left\|\boldsymbol{\lambda} - \boldsymbol{z}_0\right\|^2,$$

and any fixed $\boldsymbol{x}$ supplies a finite upper bound. The equality-only case of Theorem 19.1 therefore applies to this auxiliary problem. Its dual function, with multiplier $\boldsymbol{z}$, is

$$\begin{aligned}&\inf_{\boldsymbol{x},\boldsymbol{u}}\left\{f(\boldsymbol{x}) + \boldsymbol{\lambda}^{\mathsf{T}}\boldsymbol{u} + \frac{\rho}{2}\left\|\boldsymbol{u}\right\|^2 + \boldsymbol{z}^{\mathsf{T}}(\boldsymbol{B}\boldsymbol{x} - \boldsymbol{b} - \boldsymbol{u})\right\} \\ &\quad= \theta(\boldsymbol{z}) + \inf_{\boldsymbol{u}}\left\{(\boldsymbol{\lambda} - \boldsymbol{z})^{\mathsf{T}}\boldsymbol{u} + \frac{\rho}{2}\left\|\boldsymbol{u}\right\|^2\right\} \\ &\quad= \theta(\boldsymbol{z}) - \frac{1}{2\rho}\left\|\boldsymbol{z} - \boldsymbol{\lambda}\right\|^2.\end{aligned}$$

Thus the theorem, applied to this auxiliary equality problem, proves

$$\theta_\rho(\boldsymbol{\lambda}) = \inf_{\boldsymbol{x}} L_A(\boldsymbol{x}, \boldsymbol{\lambda}; \rho) \qquad (\boldsymbol{\lambda} \in \mathbb{R}^m). \tag{19.24}$$

This argument supplies the value identity without assuming an optimizer of either the original primal problem or the augmented subproblem.

At a center where the augmented minimum is attained, the earlier regularized-saddle calculation identifies its residual as well:

$$\begin{aligned} T_\rho(\boldsymbol{\lambda}) &= \boldsymbol{\lambda} + \rho \boldsymbol{g}(\boldsymbol{x}_\rho(\boldsymbol{\lambda})), \\ \nabla \theta_\rho(\boldsymbol{\lambda}) &= \boldsymbol{g}(\boldsymbol{x}_\rho(\boldsymbol{\lambda})), \end{aligned} \tag{19.25}$$

where $\boldsymbol{x}_\rho(\boldsymbol{\lambda})$ is any global minimizer. The primal minimizer may be nonunique, but all choices have the same residual because $T_\rho(\boldsymbol{\lambda})$ is unique. If the augmented infimum is not attained, the envelope lemma still applies to a proper closed dual; it does not by itself supply a primal minimizer for computing this residual. Thus exact ALM also has the explicit gradient form

$$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho \nabla \theta_\rho(\boldsymbol{\lambda}^{(k)}). \tag{19.26}$$

The same iteration is implicit ascent on the possibly nonsmooth ordinary dual and explicit gradient ascent on its smooth envelope.

We should also check that smoothing has not changed the optimization target. Choosing $\boldsymbol{z} = \boldsymbol{\lambda}$ gives $\theta_\rho(\boldsymbol{\lambda}) \geq \theta(\boldsymbol{\lambda})$, while subtracting a nonnegative quadratic gives $\theta_\rho(\boldsymbol{\lambda}) \leq \sup \theta$. Taking suprema yields

$$\sup_{\boldsymbol{\lambda}} \theta_\rho(\boldsymbol{\lambda}) = \sup_{\boldsymbol{\lambda}} \theta(\boldsymbol{\lambda}).$$

If $\boldsymbol{\lambda}$ maximizes θ, then it uniquely solves its own proximal problem, so $T_\rho(\boldsymbol{\lambda}) = \boldsymbol{\lambda}$ and $\theta_\rho(\boldsymbol{\lambda}) = \theta(\boldsymbol{\lambda})$. Conversely, if $\boldsymbol{\lambda}$ maximizes the differentiable concave envelope, its gradient vanishes. Hence $T_\rho(\boldsymbol{\lambda}) = \boldsymbol{\lambda}$, and the supergradient inequality in the proof of Lemma 19.7 becomes $\theta(\boldsymbol{z}) \leq \theta(\boldsymbol{\lambda})$ for every $\boldsymbol{z}$. Therefore

$$\arg\max \theta_\rho = \arg\max \theta. \tag{19.27}$$

There is no need to send ρ to zero to recover the dual maximizers. This differs from smoothing schemes whose regularized problem has a different optimizer.

For a simple picture, take the generic concave function $\theta(\lambda) = -|\lambda|$. Maximizing separately on $z \geq 0$ and $z \leq 0$ gives

$$T_\rho(\lambda) = \operatorname{sign}(\lambda) \max\{|\lambda| - \rho, 0\},$$

$$\theta_\rho(\lambda) = \begin{cases} -\lambda^2/(2\rho), & |\lambda| \leq \rho, \\ -|\lambda| + \rho/2, & |\lambda| \geq \rho. \end{cases}$$

The slopes of the two formulas agree at $\lambda = \pm\rho$, and the corner at zero has become a quadratic segment. Both functions still have the same unique maximizer, zero. See Figure 19.4.

The proximal map's continuity also completes the exact multiplier convergence argument. Under dual attainment, Equation (19.19) makes the multiplier sequence bounded and its increments tend to zero. Choose a convergent subsequence with limit $\bar{\boldsymbol{\lambda}}$. Since $\boldsymbol{\lambda}^{(k+1)} = T_\rho(\boldsymbol{\lambda}^{(k)})$, continuity gives $T_\rho(\bar{\boldsymbol{\lambda}}) = \bar{\boldsymbol{\lambda}}$, so the limit is dual optimal. Applying Equation (19.19) with this particular optimizer shows that all subsequent distances to it are nonincreasing. The subsequence of these distances tends to zero, so the entire sequence converges to $\bar{\boldsymbol{\lambda}}$. This is the precise fixed-ρ conclusion under the convex assumptions and exact attained inner solves.

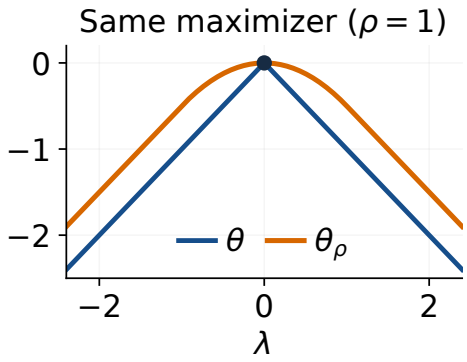


Figure 19.4: A generic illustration of the concave envelope: the kinked function $\theta(\lambda) = -|\lambda|$ and its smooth envelope have the same maximizer. The smoothing occurs after maximizing over the auxiliary multiplier, not by merely subtracting a quadratic from θ.

19.7.6 ALM Inner accuracy

The preceding interpretation identifies both a benefit and a computational cost. The augmented dual is smooth even when the ordinary dual is not, but one gradient evaluation in Equation (19.25) requires solving an augmented primal problem. Increasing ρ reduces the bound $1/\rho$ on the gradient's Lipschitz constant while weakening the dual proximal penalty and potentially worsening the inner conditioning. Moreover, if f separates across primal blocks, a term such as $\|\sum_i \boldsymbol{B}_i \boldsymbol{x}_i - \boldsymbol{b}\|^2$ generally couples them again. The ordinary Lagrangian may retain a useful separation that ALM loses.

Chapter 15 implements inexact inner solves. If $\widetilde{\boldsymbol{x}}$ is the computed point and $\widetilde{\boldsymbol{\lambda}}^+ = \boldsymbol{\lambda}^{(k)} + \rho \boldsymbol{g}(\widetilde{\boldsymbol{x}})$, then the remaining inner gradient is exactly

$$\boldsymbol{e} = \nabla_{\boldsymbol{x}} L_A(\widetilde{\boldsymbol{x}}, \boldsymbol{\lambda}^{(k)}; \rho) = \nabla_{\boldsymbol{x}} L(\widetilde{\boldsymbol{x}}, \widetilde{\boldsymbol{\lambda}}^+).$$

When $\boldsymbol{e} \neq \boldsymbol{0}$, the argument that the computed point globally minimizes this ordinary Lagrangian no longer applies. The update is an approximate proximal or envelope-gradient step, rather than the exact step used in the Fejér proof. This changes the algorithmic approximation, not the mathematical definition or smoothness of θ_ρ.

For example, suppose the augmented subproblem is globally strongly convex with known modulus $a > 0$. Strong monotonicity of its gradient and Cauchy–Schwarz give $\|\widetilde{\boldsymbol{x}} - \boldsymbol{x}_\rho\| \leq \|\boldsymbol{e}\| / a$. Consequently, using the Euclidean operator norm of $\boldsymbol{B}$,

$$\left\|\widetilde{\boldsymbol{\lambda}}^+ - T_\rho(\boldsymbol{\lambda}^{(k)})\right\| \leq \frac{\rho \|\boldsymbol{B}\|}{a} \|\boldsymbol{e}\| \leq \frac{\rho \|\boldsymbol{B}\| \sqrt{n}}{a} \varepsilon_{\text{in}}^{(k)}$$

whenever $\|\boldsymbol{e}\|_\infty \leq \varepsilon_{\text{in}}^{(k)}$. This illustrates how an inner tolerance enters an outer error estimate; without such a curvature bound, a small gradient alone does not provide this estimate. The objective-value correction needed to certify an inexact inner value was derived in Equation (18.21). A fixed positive inner tolerance is a practical implementation choice, but by itself does not establish the exact convergence conclusion above.

Convexity and affinity are also substantive assumptions. For nonconvex data, inner stationarity need not globally minimize the ordinary Lagrangian at the corrected multiplier. The regularized saddle argument and the resulting identification with the ordinary dual envelope can then fail. Although a proper ordinary dual remains closed and concave and therefore has a smooth envelope, a nonconvex ALM step need not evaluate it. Chapter 15's local analysis, including its sufficiently large penalty requirement, addresses a different set of hypotheses.

To prepare for Chapter 20, suppose instead that an ordinary Lagrangian minimizer exists at the current multiplier:

$$\boldsymbol{x}(\boldsymbol{\lambda}) \in \arg\min_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\lambda}).$$

The same evaluation argument as in Equation (19.18) now supplies a supergradient at $\boldsymbol{\lambda}$ itself:

$$\theta(\boldsymbol{z}) \le \theta(\boldsymbol{\lambda}) + \boldsymbol{g}(\boldsymbol{x}(\boldsymbol{\lambda}))^{\mathsf{T}}(\boldsymbol{z} - \boldsymbol{\lambda}).$$

This motivates the explicit dual update

$$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \alpha^{(k)} \boldsymbol{g}(\boldsymbol{x}(\boldsymbol{\lambda}^{(k)})).$$

It uses the ordinary inner objective L and works directly with the possibly nonsmooth θ, whereas ALM uses L_A and follows the gradient of θ_ρ. At a kink, a chosen supergradient need not be an ascent direction, so step lengths and possible oscillation require their own analysis. Chapter 20 develops these issues, together with inner accuracy, primal recovery, and projection onto $\mathbb{R}^p_+$ when inequality multipliers are included. The distinction is between the inner problems and the dual information they supply, not a general claim that one method is always faster.

19.8 Supplement: proof of convex strong duality

We now prove Theorem 19.1 for the full-space formulation Equation (19.3). Thus $f, h_1, \dots, h_p$ are finite convex functions on $\mathbb{R}^n$, p^* is finite, and a point $\widehat{\boldsymbol{x}}$ satisfies $\boldsymbol{B}\widehat{\boldsymbol{x}} = \boldsymbol{b}$ and $\boldsymbol{h}(\widehat{\boldsymbol{x}}) < \boldsymbol{0}$. The proof first removes redundant equality equations. It then separates a convex set of achievable perturbations from an unattainably low objective value. Strict feasibility makes the objective coefficient of that separating hyperplane positive, allowing us to normalize it into Lagrange multipliers.

Lemma 19.8 (Reduction of redundant equalities)**.** *Let $\widehat{\boldsymbol{x}} \in \mathbb{R}^n$ satisfy $\boldsymbol{B}\widehat{\boldsymbol{x}} = \boldsymbol{b}$ and put $r = \operatorname{rank} \boldsymbol{B}$. There is a linear map $\boldsymbol{M} : \mathbb{R}^m \to \mathbb{R}^r$ such that*

$$\widetilde{\boldsymbol{B}}\boldsymbol{x} = \widetilde{\boldsymbol{b}}, \qquad \widetilde{\boldsymbol{B}} = \boldsymbol{M}\boldsymbol{B}, \quad \widetilde{\boldsymbol{b}} = \boldsymbol{M}\boldsymbol{b},$$

is equivalent to $\boldsymbol{B}\boldsymbol{x} = \boldsymbol{b}$ and $\widetilde{\boldsymbol{B}}$ has full row rank. A reduced multiplier $\widetilde{\boldsymbol{\lambda}}$ induces an original multiplier $\boldsymbol{\lambda} = \boldsymbol{M}^{\mathsf{T}}\widetilde{\boldsymbol{\lambda}}$ with the same Lagrangian term.

Proof. Choose $\boldsymbol{M}$ whose restriction to $\operatorname{Range}(\boldsymbol{B})$ is an isomorphism onto $\mathbb{R}^r$. For every $\boldsymbol{x}$,

$$\boldsymbol{B}\boldsymbol{x} - \boldsymbol{b} = \boldsymbol{B}(\boldsymbol{x} - \widehat{\boldsymbol{x}}) \in \operatorname{Range}(\boldsymbol{B}).$$

Injectivity on this range gives $\boldsymbol{M}(\boldsymbol{B}\boldsymbol{x} - \boldsymbol{b}) = \boldsymbol{0}$ if and only if $\boldsymbol{B}\boldsymbol{x} - \boldsymbol{b} = \boldsymbol{0}$. Surjectivity gives $\operatorname{Range}(\widetilde{\boldsymbol{B}}) = \mathbb{R}^r$. Finally,

$$\widetilde{\boldsymbol{\lambda}}^{\mathsf{T}}(\widetilde{\boldsymbol{B}}\boldsymbol{x} - \widetilde{\boldsymbol{b}}) = (\boldsymbol{M}^{\mathsf{T}}\widetilde{\boldsymbol{\lambda}})^{\mathsf{T}}(\boldsymbol{B}\boldsymbol{x} - \boldsymbol{b}),$$

which proves the multiplier claim. This includes $r = 0$, when the reduced system has no equality constraints. □

After applying this reduction and later mapping the multiplier back, we may assume without loss of generality that

$$\operatorname{Range}(\boldsymbol{B}) = \mathbb{R}^m. \tag{19.28}$$

Lemma 19.9 (Slater separation)**.** *Under the preceding convexity, finiteness, strict-feasibility, and full-row-rank assumptions, there are* $\bar{\boldsymbol{\lambda}} \in \mathbb{R}^m$, $\bar{\boldsymbol{\nu}} \geq \mathbf{0}$, *and* $\eta > 0$ *such that*

$$\eta f(\boldsymbol{x}) + \bar{\boldsymbol{\lambda}}^{\mathsf{T}}(\boldsymbol{B}\boldsymbol{x} - \boldsymbol{b}) + \bar{\boldsymbol{\nu}}^{\mathsf{T}}\boldsymbol{h}(\boldsymbol{x}) \geq \eta p^* \qquad (\boldsymbol{x} \in \mathbb{R}^n). \tag{19.29}$$

Proof. Define the strict upper perturbation set

$$\mathcal{O} = \left\{ (\boldsymbol{u}, \boldsymbol{v}, t) : \begin{array}{l} \text{there exists } \boldsymbol{x} \in \mathbb{R}^n \text{ with} \\ \boldsymbol{B}\boldsymbol{x} - \boldsymbol{b} = \boldsymbol{u}, \quad \boldsymbol{h}(\boldsymbol{x}) < \boldsymbol{v}, \quad f(\boldsymbol{x}) < t \end{array} \right\}.$$

To verify convexity, take witnesses $\boldsymbol{x}_1, \boldsymbol{x}_2$ for two elements of $\mathcal{O}$. Their convex combination gives the corresponding equality perturbation by affinity of $\boldsymbol{B}\boldsymbol{x} - \boldsymbol{b}$. Convexity of f and every h_j preserves the strict upper inequalities. Thus $\mathcal{O}$ is convex.

We next show that $\mathcal{O}$ is open. Fix $(\boldsymbol{u}, \boldsymbol{v}, t) \in \mathcal{O}$ with witness $\boldsymbol{x}$. Full row rank gives a linear right inverse $\boldsymbol{R} : \mathbb{R}^m \to \mathbb{R}^n$ with $\boldsymbol{B}\boldsymbol{R} = \boldsymbol{I}_m$. All finite convex functions on $\mathbb{R}^n$ are continuous. Define the positive margins

$$\gamma_h = \begin{cases} \min_j\{v_j - h_j(\boldsymbol{x})\}, & p > 0, \\ 1, & p = 0, \end{cases} \qquad \gamma_f = t - f(\boldsymbol{x}) > 0.$$

Continuity gives $\varepsilon > 0$ such that, whenever $\|\Delta\boldsymbol{u}\| < \varepsilon$, the point $\boldsymbol{x}' = \boldsymbol{x} + \boldsymbol{R}\Delta\boldsymbol{u}$ satisfies

$$|h_j(\boldsymbol{x}') - h_j(\boldsymbol{x})| < \gamma_h/3, \qquad |f(\boldsymbol{x}') - f(\boldsymbol{x})| < \gamma_f/3.$$

Shrink ε so that $\|\Delta\boldsymbol{v}\|_\infty < \varepsilon$ implies $\|\Delta\boldsymbol{v}\|_\infty < \gamma_h/3$, and $|\Delta t| < \varepsilon$ implies $|\Delta t| < \gamma_f/3$. Then

$$\boldsymbol{B}\boldsymbol{x}' - \boldsymbol{b} = \boldsymbol{u} + \Delta\boldsymbol{u}, \quad \boldsymbol{h}(\boldsymbol{x}') < \boldsymbol{v} + \Delta\boldsymbol{v}, \quad f(\boldsymbol{x}') < t + \Delta t.$$

Thus all sufficiently small perturbations stay in $\mathcal{O}$. When $p = 0$ the inequality-margin conditions are omitted; when $m = 0$ there is no equality perturbation to correct.

The point $(\mathbf{0}, \mathbf{0}, p^*)$ is outside $\mathcal{O}$: a witness would be feasible with objective below p^*. Separating this point from the nonempty open convex set gives a nonzero triple $(\bar{\boldsymbol{\lambda}}, \bar{\boldsymbol{\nu}}, \eta)$ with

$$\bar{\boldsymbol{\lambda}}^{\mathsf{T}}\boldsymbol{u} + \bar{\boldsymbol{\nu}}^{\mathsf{T}}\boldsymbol{v} + \eta t \geq \eta p^* \qquad ((\boldsymbol{u}, \boldsymbol{v}, t) \in \mathcal{O}). \tag{19.30}$$

The set is closed under increasing any coordinate of $\boldsymbol{v}$ or increasing t. If any coefficient of those coordinates were negative, sending that coordinate to $+\infty$ would contradict the inequality. Therefore $\bar{\boldsymbol{\nu}} \geq \mathbf{0}$ and $\eta \geq 0$.

To show $\eta > 0$, suppose instead that $\eta = 0$. Use the strictly feasible point $\widehat{\boldsymbol{x}}$ and let $\boldsymbol{v} \downarrow \boldsymbol{h}(\widehat{\boldsymbol{x}})$ in Equation (19.30). Since its equality residual is zero, this gives $\bar{\boldsymbol{\nu}}^{\mathsf{T}}\boldsymbol{h}(\widehat{\boldsymbol{x}}) \geq 0$. Strict negativity of every $h_j(\widehat{\boldsymbol{x}})$ forces $\bar{\boldsymbol{\nu}} = \mathbf{0}$. For every sufficiently small $\boldsymbol{u}$, the point $\widehat{\boldsymbol{x}} + \boldsymbol{R}\boldsymbol{u}$ remains strictly inequality feasible by continuity. Choose upper coordinates $\boldsymbol{v}, t$ to obtain an element of $\mathcal{O}$ with this equality residual. The separating inequality now says $\bar{\boldsymbol{\lambda}}^{\mathsf{T}}\boldsymbol{u} \geq 0$ for both signs of each small $\boldsymbol{u}$. Hence $\bar{\boldsymbol{\lambda}} = \mathbf{0}$, contradicting that the separating triple is nonzero. This argument also covers empty equality or inequality blocks.

Finally, for any $\boldsymbol{x} \in \mathbb{R}^n$, substitute

$$(\boldsymbol{u}, \boldsymbol{v}, t) = (\boldsymbol{B}\boldsymbol{x} - \boldsymbol{b}, \boldsymbol{h}(\boldsymbol{x}) + \varepsilon\mathbf{1}, f(\boldsymbol{x}) + \varepsilon)$$

into Equation (19.30) and let $\varepsilon \downarrow 0$. The result is Equation (19.29). □

19.8.1 Completion of the strong-duality proof

Proof of Theorem 19.1. Apply Lemma 19.9 and define

$$\widetilde{\boldsymbol{\lambda}} = \bar{\boldsymbol{\lambda}}/\eta, \qquad \widetilde{\boldsymbol{\nu}} = \bar{\boldsymbol{\nu}}/\eta \geq \mathbf{0}.$$

Dividing Equation (19.29) by η gives, for every $\boldsymbol{x}$,

$$f(\boldsymbol{x}) + \widetilde{\boldsymbol{\lambda}}^{\mathsf{T}}(\boldsymbol{B}\boldsymbol{x} - \boldsymbol{b}) + \widetilde{\boldsymbol{\nu}}^{\mathsf{T}}\boldsymbol{h}(\boldsymbol{x}) \geq p^*.$$

Taking the infimum over $\mathbb{R}^n$ gives $\theta(\widetilde{\boldsymbol{\lambda}}, \widetilde{\boldsymbol{\nu}}) \geq p^*$, while weak duality gives the reverse inequality. Thus

$$\theta(\widetilde{\boldsymbol{\lambda}}, \widetilde{\boldsymbol{\nu}}) = p^* = d^*.$$

If equalities were reduced, use Lemma 19.8 to map $\widetilde{\boldsymbol{\lambda}}$ back to the original equality coordinates. The Lagrangian values are unchanged, so the original dual also attains p^*. □

19.8.2 Supplement: multiplier sensitivity

Continue under the hypotheses of Theorem 19.1, and consider the convex program in Equation (19.3). Define the extended-real perturbation value function

$$v(\boldsymbol{u}, \boldsymbol{v}) \coloneqq \inf\left\{ f(\boldsymbol{x}) : \boldsymbol{x} \in \mathbb{R}^n,\ \boldsymbol{B}\boldsymbol{x} - \boldsymbol{b} = \boldsymbol{u},\quad \boldsymbol{h}(\boldsymbol{x}) \leq \boldsymbol{v} \right\}. \tag{19.31}$$

We use $v(\boldsymbol{u}, \boldsymbol{v}) = +\infty$ if the perturbed problem is infeasible. Thus $v(\mathbf{0}, \mathbf{0}) = p^*$. The sign convention is important: $\boldsymbol{B}\boldsymbol{x} - \boldsymbol{b} = \boldsymbol{u}$ means that the equality right-hand side has changed from $\boldsymbol{b}$ to $\boldsymbol{b} + \boldsymbol{u}$, whereas increasing $\boldsymbol{v}$ relaxes the inequalities.

Proposition 19.10 (Convexity of the perturbation value)**.** *Under the convex assumptions of Equation (19.3), the function $(\boldsymbol{u}, \boldsymbol{v}) \mapsto v(\boldsymbol{u}, \boldsymbol{v})$ is convex as an extended-real function.*

Proof. Let $(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ be the dual optimizer guaranteed by Theorem 19.1. The same weak-duality calculation applied to a perturbed feasible point gives

$$v(\boldsymbol{u}, \boldsymbol{v}) \geq p^* - (\boldsymbol{\lambda}^*)^{\mathsf{T}}\boldsymbol{u} - (\boldsymbol{\nu}^*)^{\mathsf{T}}\boldsymbol{v}.$$

Thus v never takes the value $-\infty$. Take perturbations $\boldsymbol{z}_i = (\boldsymbol{u}_i, \boldsymbol{v}_i)$ with finite values, fix $t \in (0, 1)$, and let $\delta > 0$. By the definition of an infimum, there are points $\boldsymbol{x}_i \in \mathbb{R}^n$ feasible for $\boldsymbol{z}_i$ such that

$$f(\boldsymbol{x}_i) \leq v(\boldsymbol{z}_i) + \delta, \qquad i = 1, 2.$$

For $\boldsymbol{x}_t = t\boldsymbol{x}_1 + (1-t)\boldsymbol{x}_2$, affinity of the equality map and convexity of every h_j give

$$\boldsymbol{B}\boldsymbol{x}_t - \boldsymbol{b} = t\boldsymbol{u}_1 + (1-t)\boldsymbol{u}_2, \qquad \boldsymbol{h}(\boldsymbol{x}_t) \leq t\boldsymbol{v}_1 + (1-t)\boldsymbol{v}_2.$$

Convexity of f therefore yields

$$v(t\boldsymbol{z}_1 + (1-t)\boldsymbol{z}_2) \leq f(\boldsymbol{x}_t) \leq tv(\boldsymbol{z}_1) + (1-t)v(\boldsymbol{z}_2) + \delta.$$

Let $\delta \downarrow 0$. If either endpoint value is $+\infty$, the convexity inequality is automatic. This proves the claim. □

For a convex extended-real function v that is finite at $\boldsymbol{z}_0$, its subdifferential is

$$\partial v(\boldsymbol{z}_0) = \left\{ \boldsymbol{s} : v(\boldsymbol{z}) \geq v(\boldsymbol{z}_0) + \boldsymbol{s}^{\mathsf{T}}(\boldsymbol{z} - \boldsymbol{z}_0) \text{ for every } \boldsymbol{z} \right\}.$$

Let $\mathcal{M}^*$ denote the set of dual-optimal multiplier pairs:

$$\mathcal{M}^* := \{(\boldsymbol{\lambda}, \boldsymbol{\nu}) : \boldsymbol{\nu} \geq \boldsymbol{0},\ \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) = p^*\}.$$

This set is nonempty under the hypotheses of Theorem 19.1.

Theorem 19.11 (Optimal multipliers are signed value-function subgradients)**.** *Suppose $p^* = v(\boldsymbol{0}, \boldsymbol{0})$ is finite. Then*

$$\partial v(\boldsymbol{0}, \boldsymbol{0}) = \{(-\boldsymbol{\lambda}, -\boldsymbol{\nu}) : (\boldsymbol{\lambda}, \boldsymbol{\nu}) \in \mathcal{M}^*\}. \tag{19.32}$$

Consequently, an optimal multiplier supports the perturbation value globally:

$$v(\boldsymbol{u}, \boldsymbol{v}) \geq p^* - \boldsymbol{\lambda}^{\mathsf{T}}\boldsymbol{u} - \boldsymbol{\nu}^{\mathsf{T}}\boldsymbol{v} \qquad \text{for every } (\boldsymbol{u}, \boldsymbol{v}). \tag{19.33}$$

Proof. First take $(\boldsymbol{\lambda}, \boldsymbol{\nu}) \in \mathcal{M}^*$. If $\boldsymbol{x}$ is feasible for the perturbation $(\boldsymbol{u}, \boldsymbol{v})$, then

$$p^* = \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\boldsymbol{x}) + \boldsymbol{\lambda}^{\mathsf{T}}\boldsymbol{u} + \boldsymbol{\nu}^{\mathsf{T}}\boldsymbol{h}(\boldsymbol{x}) \leq f(\boldsymbol{x}) + \boldsymbol{\lambda}^{\mathsf{T}}\boldsymbol{u} + \boldsymbol{\nu}^{\mathsf{T}}\boldsymbol{v}.$$

Taking the infimum over all such $\boldsymbol{x}$ proves Equation (19.33); it is exactly the subgradient inequality for $(-\boldsymbol{\lambda}, -\boldsymbol{\nu})$ at the origin.

Conversely, write $\boldsymbol{s} = (\boldsymbol{s}_u, \boldsymbol{s}_v) \in \partial v(\boldsymbol{0}, \boldsymbol{0})$. For every $t > 0$, relaxing only the j-th inequality cannot increase the value, so

$$p^* + t(\boldsymbol{s}_v)_j \leq v(\boldsymbol{0}, t\boldsymbol{e}_j) \leq p^*.$$

Hence $\boldsymbol{s}_v \leq \boldsymbol{0}$. Set $\boldsymbol{\lambda} = -\boldsymbol{s}_u$ and $\boldsymbol{\nu} = -\boldsymbol{s}_v \geq \boldsymbol{0}$. Given any $\boldsymbol{x} \in \mathbb{R}^n$, the same point is feasible for the perturbation $(\boldsymbol{B}\boldsymbol{x} - \boldsymbol{b}, \boldsymbol{h}(\boldsymbol{x}))$. Therefore

$$\begin{aligned} f(\boldsymbol{x}) &\geq v(\boldsymbol{B}\boldsymbol{x} - \boldsymbol{b}, \boldsymbol{h}(\boldsymbol{x})) \\ &\geq p^* + \boldsymbol{s}_u^{\mathsf{T}}(\boldsymbol{B}\boldsymbol{x} - \boldsymbol{b}) + \boldsymbol{s}_v^{\mathsf{T}}\boldsymbol{h}(\boldsymbol{x}), \end{aligned}$$

or equivalently $L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \geq p^*$. Taking the infimum over $\boldsymbol{x} \in \mathbb{R}^n$ gives $\theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \geq p^*$. Weak duality gives the reverse inequality, so $(\boldsymbol{\lambda}, \boldsymbol{\nu}) \in \mathcal{M}^*$. □

The theorem is already a rigorous global sensitivity statement. Calling a multiplier *the derivative* of the optimal value requires more regularity, because a convex value function may be nonsmooth or finite only on a lower-dimensional set.

Theorem 19.12 (Directional sensitivity under local finiteness)**.** *In addition to the hypotheses of Theorem 19.11, suppose that v is finite on an open neighbourhood of $(\boldsymbol{0}, \boldsymbol{0})$. Then, for every perturbation direction $(\boldsymbol{d}_u, \boldsymbol{d}_v)$, the one-sided directional derivative exists and satisfies*

$$v'((\boldsymbol{0}, \boldsymbol{0}); (\boldsymbol{d}_u, \boldsymbol{d}_v)) = \max_{(\boldsymbol{\lambda}, \boldsymbol{\nu}) \in \mathcal{M}^*} \left\{ -\boldsymbol{\lambda}^{\mathsf{T}}\boldsymbol{d}_u - \boldsymbol{\nu}^{\mathsf{T}}\boldsymbol{d}_v \right\}. \tag{19.34}$$

The maximum is attained.

Proof. We first prove the standard local fact needed here. Let w be a convex function finite on a neighbourhood of the origin. It is continuous there, has a nonempty subdifferential at every nearby point by the supporting-hyperplane theorem, and its nearby subgradients are bounded. For the last claim, choose $r > 0$ such that w is finite on $B(\mathbf{0}, r)$. Continuity bounds w in absolute value by some M on $B(\mathbf{0}, r/2)$. If $\boldsymbol{z} \in B(\mathbf{0}, r/4)$, $\boldsymbol{s} \in \partial w(\boldsymbol{z})$, and $\boldsymbol{e}$ is a unit vector, apply the subgradient inequality at $\boldsymbol{z} \pm (r/4)\boldsymbol{e}$ to obtain $|\boldsymbol{s}^{\mathsf{T}}\boldsymbol{e}| \le 8M/r$. Thus the nearby subgradients are uniformly bounded.

Fix a direction $\boldsymbol{d}$. Convexity of the one-variable function $t \mapsto w(t\boldsymbol{d})$ shows that

$$q(\boldsymbol{d}) \coloneqq \lim_{t \downarrow 0} \frac{w(t\boldsymbol{d}) - w(\mathbf{0})}{t}$$

exists. Every $\boldsymbol{s} \in \partial w(\mathbf{0})$ gives $q(\boldsymbol{d}) \ge \boldsymbol{s}^{\mathsf{T}}\boldsymbol{d}$. Conversely, take $t_\ell \downarrow 0$ and $\boldsymbol{s}_\ell \in \partial w(t_\ell \boldsymbol{d})$. The preceding boundedness permits a subsequence with $\boldsymbol{s}_\ell \to \bar{\boldsymbol{s}}$. Passing to the limit in the subgradient inequalities shows $\bar{\boldsymbol{s}} \in \partial w(\mathbf{0})$, while the subgradient inequality between $t_\ell \boldsymbol{d}$ and $\mathbf{0}$ gives

$$\frac{w(t_\ell \boldsymbol{d}) - w(\mathbf{0})}{t_\ell} \le \boldsymbol{s}_\ell^{\mathsf{T}}\boldsymbol{d}.$$

Therefore

$$q(\boldsymbol{d}) = \max_{\boldsymbol{s} \in \partial w(\mathbf{0})} \boldsymbol{s}^{\mathsf{T}}\boldsymbol{d}.$$

The same boundedness and closedness show that the subdifferential is compact, so the maximum is attained. Apply this fact to $w(\boldsymbol{z}) = v(\boldsymbol{u}, \boldsymbol{v})$, then substitute the subdifferential identity in Equation (19.32). □

Corollary 19.13 (One-dimensional slopes and differentiability)**.** *Under the hypotheses of Theorem 19.12, set $\psi(t) = v(t\boldsymbol{d}_u, t\boldsymbol{d}_v)$ and*

$$\sigma(\boldsymbol{\lambda}, \boldsymbol{\nu}) = -\boldsymbol{\lambda}^{\mathsf{T}}\boldsymbol{d}_u - \boldsymbol{\nu}^{\mathsf{T}}\boldsymbol{d}_v.$$

Then

$$\psi'_+(0) = \max_{(\boldsymbol{\lambda}, \boldsymbol{\nu}) \in \mathcal{M}^*} \sigma(\boldsymbol{\lambda}, \boldsymbol{\nu}), \qquad \psi'_-(0) = \min_{(\boldsymbol{\lambda}, \boldsymbol{\nu}) \in \mathcal{M}^*} \sigma(\boldsymbol{\lambda}, \boldsymbol{\nu}).$$

If the optimal multiplier pair is unique, then v is differentiable at the origin and

$$\nabla v(\mathbf{0}, \mathbf{0}) = (-\boldsymbol{\lambda}^*, -\boldsymbol{\nu}^*).$$

Thus increasing an inequality right-hand side has first-order slope $-\nu_j^* \le 0$, while an equality right-hand-side perturbation has slope $-\lambda_i^*$. Multiple optimal multipliers can instead produce a kink and a set of valid supporting slopes.

Proof. The formula for the right derivative is Equation (19.34). Applying the same formula to the direction $(-\boldsymbol{d}_u, -\boldsymbol{d}_v)$ and using $\psi'_-(0) = -v'((\mathbf{0}, \mathbf{0}); (-\boldsymbol{d}_u, -\boldsymbol{d}_v))$ gives the left derivative. If $\mathcal{M}^*$ is a singleton, then Equation (19.32) makes $\partial v(\mathbf{0}, \mathbf{0})$ a singleton. A convex function finite on an open neighbourhood is differentiable at an interior point exactly when its subdifferential there is a singleton; the unique subgradient is its gradient. The signs follow directly from the definition of the perturbations in Equation (19.31). □

Warning

The multiplier–subgradient identity is valid without differentiability, but a literal "multiplier equals the marginal derivative" statement is not. Before using that language, one must fix the perturbation sign convention and verify local finiteness (or another regularity condition ensuring directional sensitivity). At a kink there is a set of slopes; with redundant equalities the value function may be finite only on a proper affine subspace.

19.8.3 Supplement: compact convex–concave minimax

The universal minimax inequality used at the start of this chapter says $\sup\inf \le \inf\sup$. Equality can follow from hypotheses that look quite different from Slater's condition. The following finite-dimensional version is useful for comparison.

Theorem 19.14 (Compact convex–concave minimax)**.** *Let $X \subseteq \mathbb{R}^n$ and $Y \subseteq \mathbb{R}^q$ be nonempty compact convex sets. Suppose $\Phi : X \times Y \to \mathbb{R}$ is continuous, $\Phi(\cdot, \boldsymbol{y})$ is convex on X for every $\boldsymbol{y} \in Y$, and $\Phi(\boldsymbol{x}, \cdot)$ is concave on Y for every $\boldsymbol{x} \in X$. Then*

$$\min_{\boldsymbol{x}\in X}\max_{\boldsymbol{y}\in Y}\Phi(\boldsymbol{x},\boldsymbol{y}) = \max_{\boldsymbol{y}\in Y}\min_{\boldsymbol{x}\in X}\Phi(\boldsymbol{x},\boldsymbol{y}), \tag{19.35}$$

and both outer extrema are attained.

Proof. Compactness and continuity imply that

$$F(\boldsymbol{x}) = \max_{\boldsymbol{y}\in Y}\Phi(\boldsymbol{x},\boldsymbol{y}), \qquad G(\boldsymbol{y}) = \min_{\boldsymbol{x}\in X}\Phi(\boldsymbol{x},\boldsymbol{y})$$

are continuous: uniform continuity of Φ on the compact product controls the change in either optimized value. Hence $\alpha = \min_X F$ and $\beta = \max_Y G$ are attained. The minimax inequality gives $\beta \le \alpha$.

Fix $c < \alpha$, and for $\boldsymbol{y} \in Y$ define the closed convex subset

$$K_{\boldsymbol{y}} = \{\boldsymbol{x} \in X : \Phi(\boldsymbol{x},\boldsymbol{y}) \le c\}.$$

Their total intersection is empty, for otherwise some $\boldsymbol{x}$ would satisfy $F(\boldsymbol{x}) \le c < \alpha$. Because these are closed subsets of the compact set X, the finite-intersection property provides $\boldsymbol{y}_1, \dots, \boldsymbol{y}_N \in Y$ such that $\bigcap_{i=1}^N K_{\boldsymbol{y}_i} = \varnothing$. Consequently,

$$\alpha_N \coloneqq \min_{\boldsymbol{x}\in X}\max_{1\le i\le N}\Phi(\boldsymbol{x},\boldsymbol{y}_i) > c.$$

To combine these finitely many functions with nonnegative weights, consider their upper image

$$C = \left\{\boldsymbol{z} \in \mathbb{R}^N : \text{there is } \boldsymbol{x} \in X \text{ with } \Phi(\boldsymbol{x},\boldsymbol{y}_i) \le z_i \ (i = 1,\dots,N)\right\}.$$

Convexity of $\Phi(\cdot, \boldsymbol{y}_i)$ and X makes C convex. It is closed: if $\boldsymbol{z}_k \to \boldsymbol{z}$ with witnesses $\boldsymbol{x}_k \in X$, compactness supplies a convergent subsequence, and continuity passes all inequalities to the limit. The point $c\mathbf{1}$ is outside C because $\alpha_N > c$. Strict separation of a point from a nonempty closed convex set gives a nonzero vector $\boldsymbol{\pi}$ with

$$\inf_{\boldsymbol{z}\in C}\boldsymbol{\pi}^{\mathsf{T}}\boldsymbol{z} > c\boldsymbol{\pi}^{\mathsf{T}}\mathbf{1}.$$

The set C is closed under adding nonnegative vectors. A negative component of $\boldsymbol{\pi}$ would make this infimum $-\infty$, so $\boldsymbol{\pi} \geq \mathbf{0}$. Normalize by its positive sum to obtain $\sum_i \pi_i = 1$. For nonnegative weights the least upper coordinates can be used, and therefore

$$\min_{\boldsymbol{x} \in X} \sum_{i=1}^{N} \pi_i \Phi(\boldsymbol{x}, \boldsymbol{y}_i) > c. \tag{19.36}$$

Let $\bar{\boldsymbol{y}} = \sum_i \pi_i \boldsymbol{y}_i \in Y$. Concavity in $\boldsymbol{y}$ gives, for every $\boldsymbol{x} \in X$,

$$\Phi(\boldsymbol{x}, \bar{\boldsymbol{y}}) \geq \sum_{i=1}^{N} \pi_i \Phi(\boldsymbol{x}, \boldsymbol{y}_i).$$

Taking the minimum over $\boldsymbol{x}$ and using Equation (19.36) yields $G(\bar{\boldsymbol{y}}) > c$. Thus $\beta > c$ for every $c < \alpha$, and therefore $\beta \geq \alpha$. Together with the minimax inequality, this proves equality. □

Warning

For a Lagrangian, compare

$$\inf_{\boldsymbol{x} \in \mathbb{R}^n} \sup_{(\boldsymbol{\lambda}, \boldsymbol{\nu}) \in \mathbb{R}^m \times \mathbb{R}^p_+} L \quad \text{with} \quad \sup_{(\boldsymbol{\lambda}, \boldsymbol{\nu}) \in \mathbb{R}^m \times \mathbb{R}^p_+} \inf_{\boldsymbol{x} \in \mathbb{R}^n} L.$$

The primal and multiplier spaces are generally noncompact, so Theorem 19.14 usually cannot be applied directly. Slater's theorem instead uses the perturbation geometry of finitely many convex constraints and permits unbounded spaces. A separate minimax argument must verify its compactness, continuity, and convex–concave hypotheses. Merely writing "inf sup = sup inf" proves neither Lagrangian strong duality nor dual attainment.

Chapter summary

- Weak duality gives $\sup \inf L \leq \inf \sup L$. Strong duality means these two orders have the same finite value; interchanging the optimization symbols requires justification.
- For convex f, h_j, affine equalities, and finite p^*, Slater's condition gives $d^* = p^*$ and dual attainment, but does not guarantee primal attainment. These sufficient hypotheses are not necessary.
- In linear programming, if either problem is feasible with a finite optimal value, both problems attain the same optimal value without strict Slater. Polyhedral attainment and Farkas-based multipliers justify the conclusion; calculating the bidual alone does not.
- Matching values at a feasible pair imply global Lagrangian minimization and complementary slackness, hence KKT for differentiable data. Convexity with affine equalities makes KKT sufficient for matching values; general nonconvex KKT conditions do not.
- Primal and dual optimizers with a common finite value are equivalent to a global Lagrangian saddle point. The full Lagrangian Hessian can be indefinite, and simultaneous or alternating descent–ascent need not converge even when a saddle point exists.

- For a convex objective and affine equalities, an attained exact ALM inner minimum gives a proximal step on the ordinary dual, equivalently

$$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho \nabla \theta_\rho(\boldsymbol{\lambda}^{(k)}).$$

  The proximal term damps the multiplier update. The envelope θ_ρ is smooth and has the same supremum and maximizers as θ; subtracting a quadratic from θ alone need not make it smooth.

- Under these convex assumptions, dual attainment and attained exact inner solves at every iteration give multiplier convergence with fixed $\rho > 0$. Larger ρ can worsen inner conditioning; inexact solves and nonconvex problems require additional analysis.

19.9 Exercises

Exercise 19.1 (Linear-programming duality)**.** Derive the dual of $\min\{2x_1+x_2 : x_1+x_2 = 1,\ \boldsymbol{x} \geq \mathbf{0}\}$. Solve both problems and identify the complementary products. Derive the bidual and explain why that calculation alone does not prove equality of values.

Exercise 19.2 (Matching values)**.** Suppose a feasible point has objective value 7 and a dual-feasible multiplier has dual value 7. Prove optimality of both points without assuming convexity. If the data are differentiable, derive each KKT condition from the value equality.

Exercise 19.3 (Attainment distinction)**.** For the unconstrained objective $f(x) = e^{-x}$ on $\mathbb{R}$, determine the primal value and its attainment status. Formulate the constraint-free dual precisely and determine its attainment status. Explain why zero gap does not give a saddle point.

Exercise 19.4 (Quadratic inequality)**.** Derive the dual of $\min_x \frac{1}{2}(x-2)^2$ subject to $x \leq 0$. Solve both problems, check Slater, and verify the saddle inequalities directly.

Exercise 19.5 (A kink in the perturbation value)**.** For $\min_{x\in\mathbb{R}} |x|$ subject to $x = 0$, perturb the equality to $x = u$. Compute $v(u)$, $\partial v(0)$, the dual function, and the complete set of dual optimizers. Verify Equation (19.32) and interpret the two one-sided derivatives of v.